

Contract for Consultant's Services

Lump-Sum

Project Name: Component 3: Tree mapping and Forestry Assessment
Grant No. 4367

Contract No. 3

Between

Serious Shea SARL

and

Sensonomic AS

Dated: 21st January 2019

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PREFACE

1. The standard Contract form consists of four parts: the Form of Contract to be signed by the Client and the Consultant, the General Conditions of Contract (GCC), including Attachment 1 (Bank's Policy – Corrupt and Fraudulent Practices); the Special Conditions of Contract (SCC); and the Appendices.
2. The General Conditions of Contract, including Attachment 1, shall not be modified. The Special Conditions of Contract that contain clauses specific to each Contract intend to supplement, but not over-write or otherwise contradict, the General Conditions.

FORM OF CONTRACT

LUMP-SUM

This CONTRACT (hereinafter called the “Contract”) is made the 26th day of the month of *November 2018*, between, on the one hand, *Serious Shea SARL* (hereinafter called the “Client”) and, on the other hand, *Sensonomic AS* (hereinafter called the “Consultant”).

WHEREAS

- (a) the Client has requested the Consultant to provide certain consulting services as defined in this Contract (hereinafter called the “Services”);
- (b) the Consultant, having represented to the Client that it has the required professional skills, expertise and technical resources, has agreed to provide the Services on the terms and conditions set forth in this Contract;
- (c) the Client has received a *grant* from the African Development Bank: toward the cost of the Services and intends to apply a portion of the proceeds of this *grant* to eligible payments under this Contract, it being understood that (i) payments by the Bank will be made only at the request of the Client and upon approval by the Bank; (ii) such payments will be subject, in all respects, to the terms and conditions of the *grant* agreement account for the purpose of any payment to persons or entities, or for any import of goods, if such payment or import, to the knowledge of the Bank, is prohibited by the decision of the United Nations Security council taken under Chapter VII of the Charter of the United Nations/; and (iii) no party other than the Client shall derive any rights from the *grant* agreement or have any claim to the *grant* proceeds;

NOW THEREFORE the parties hereto hereby agree as follows:

- 1. The following documents attached hereto shall be deemed to form an integral part of this Contract:
 - (a) The General Conditions of Contract (including Attachment 1 “Bank Policy – Corrupt and Fraudulent Practices);
 - (b) The Special Conditions of Contract;
 - (c) Appendices:
 - Appendix A: Terms of Reference
 - Appendix B: Key Experts
 - Appendix C: Breakdown of Contract Price
 - Appendix D: Form of Advance Payments Guarantee

In the event of any inconsistency between the documents, the following order of precedence shall prevail: the Special Conditions of Contract; the General Conditions of Contract, including Attachment 1; Appendix A; Appendix B; Appendix C; Appendix D. Any reference to this Contract shall include, where the context permits, a reference to its Appendices.

2. The mutual rights and obligations of the Client and the Consultant shall be as set forth in the Contract, in particular:
 - (a) the Consultant shall carry out the Services in accordance with the provisions of the Contract; and
 - (b) the Client shall make payments to the Consultant in accordance with the provisions of the Contract.

IN WITNESS WHEREOF, the Parties hereto have caused this Contract to be signed in their respective names as of the day and year first above written.

For and on behalf of Serious Shea SARL, Jon White



For and on behalf of Sensonomic AS, Anders Gundersen



GENERAL CONDITIONS OF CONTRACT

A. GENERAL PROVISIONS

1. Definitions

- 1.1. Unless the context otherwise requires, the following terms whenever used in this Contract have the following meanings:
- (a) “Applicable Guidelines” means *policy of bank details for the use of consultants which can be found at www.afdb.org*.
 - (b) “Applicable Law” means the laws and any other instruments having the force of law in the Client’s country, or in such other country as may be specified in the Special Conditions of Contract (SCC), as they may be issued and in force from time to time.
 - (c) “Bank” means the African Development Bank, the African Development Fund and the Nigeria Trust Fund as the context may require.
 - (d) “Borrower or Recipient or Beneficiary” means the Government, Government agency or other entity that signs the financing [or loan/credit/grant/project] agreement with the Bank.
 - (e) “Client” means *the implementing/ executing* agency that signs the Contract for the Services with the Selected Consultant.
 - (f) “Consultant” means a legally-established professional consulting firm or entity selected by the Client to provide the Services under the signed Contract.
 - (g) “Contract” means the legally binding written agreement signed between the Client and the Consultant and which includes all the attached documents listed in its paragraph 1 of the Form of Contract (the General Conditions (GCC), the Special Conditions (SCC), and the Appendices).
 - (h) “Day” means a working day unless indicated otherwise.
 - (i) “Effective Date” means the date on which this Contract comes into force and effect pursuant to Clause GCC 11.
 - (j) “Experts” means, collectively, Key Experts, Non-Key Experts, or any other personnel of the Consultant, Sub-consultant or JV member(s) assigned by the Consultant to perform the Services or any part thereof under the Contract.
 - (k) “Foreign Currency” means any currency other than the currency of the Client’s country.
 - (l) “GCC” means these General Conditions of Contract.
 - (m) “Government” means the government of the Client’s country.

- (n) “Joint Venture (JV)” means an association with or without a legal personality distinct from that of its members, of more than one entity where one member has the authority to conduct all businesses for and on behalf of any and all the members of the JV, and where the members of the JV are jointly and severally liable to the Client for the performance of the Contract.
- (o) “Key Expert(s)” means an individual professional whose skills, qualifications, knowledge and experience are critical to the performance of the Services under the Contract and whose Curricula Vitae (CV) was taken into account in the technical evaluation of the Consultant’s proposal.
- (p) “Local Currency” means the currency of the Client’s country.
- (q) “Non-Key Expert(s)” means an individual professional provided by the Consultant or its Sub-consultant to perform the Services or any part thereof under the Contract.
- (r) “Party” means the Client or the Consultant, as the case may be, and “Parties” means both of them.
- (s) “SCC” means the Special Conditions of Contract by which the GCC may be amended or supplemented but not over-written.
- (t) “Services” means the work to be performed by the Consultant pursuant to this Contract, as described in Appendix A hereto.
- (u) “Sub-consultants” means an entity to whom/which the Consultant subcontracts any part of the Services while remaining solely liable for the execution of the Contract.
- (v) “Third Party” means any person or entity other than the Government, the Client, the Consultant or a Sub-consultant.

2. Relationship between the Parties

- 2.1. Nothing contained herein shall be construed as establishing a relationship of master and servant or of principal and agent as between the Client and the Consultant. The Consultant, subject to this Contract, has complete charge of the Experts and Sub-consultants, if any, performing the Services and shall be fully responsible for the Services performed by them or on their behalf hereunder.

3. Law Governing Contract

- 3.1. This Contract, its meaning and interpretation, and the relation between the Parties shall be governed by the Applicable Law.

4. Language

- 4.1. This Contract has been executed in the language specified in the SCC, which shall be the binding and controlling language for all matters relating to the meaning or interpretation of this Contract.

5. Headings

- 5.1. The headings shall not limit, alter or affect the meaning of this Contract.

6. Communications

- 6.1. Any communication required or permitted to be given or made pursuant to this Contract shall be in writing in the language

specified in Clause GCC 4. Any such notice, request or consent shall be deemed to have been given or made when delivered in person to an authorized representative of the Party to whom the communication is addressed, or when sent to such Party at the address specified in the **SCC**.

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| | 6.2. | A Party may change its address for notice hereunder by giving the other Party any communication of such change to the address specified in the SCC . |
| 7. Location | 7.1. | The Services shall be performed at such locations as are specified in Appendix A hereto and, where the location of a particular task is not so specified, at such locations, whether in the Government's country or elsewhere, as the Client may approve. |
| 8. Authority of Member in Charge | 8.1. | In case the Consultant is a Joint Venture, the members hereby authorize the member specified in the SCC to act on their behalf in exercising all the Consultant's rights and obligations towards the Client under this Contract, including without limitation the receiving of instructions and payments from the Client. |
| 9. Authorized Representatives | 9.1. | Any action required or permitted to be taken, and any document required or permitted to be executed under this Contract by the Client or the Consultant may be taken or executed by the officials specified in the SCC . |
| 10. Corrupt and Fraudulent Practices | 10.1. | The Bank requires compliance with its policy in regard to corrupt and fraudulent/prohibited practices as set forth in Attachment 1 to the GCC. |
| a. Commissions and Fees | 10.2. | The Client requires the Consultant to disclose any commissions, gratuities or fees that may have been paid or are to be paid to agents or any other party with respect to the selection process or execution of the Contract. The information disclosed must include at least the name and address of the agent or other party, the amount and currency, and the purpose of the commission, gratuity or fee. Failure to disclose such commissions, gratuities or fees may result in termination of the Contract and/or sanctions by the Bank. |

B. COMMENCEMENT, COMPLETION, MODIFICATION AND TERMINATION OF CONTRACT

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| 11. Effectiveness of Contract | 11.1. | This Contract shall come into force and effect on the date (the "Effective Date") of the Client's notice to the Consultant instructing the Consultant to begin carrying out the Services. This notice shall confirm that the effectiveness conditions, if any, listed in the SCC have been met. |
| 12. Termination of Contract for Failure to Become Effective | 12.1. | If this Contract has not become effective within such time period after the date of Contract signature as specified in the SCC , either Party may, by not less than twenty two (22) days |

written notice to the other Party, declare this Contract to be null and void, and in the event of such a declaration by either Party, neither Party shall have any claim against the other Party with respect hereto.

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| 13. Commencement of Services | 13.1. | The Consultant shall confirm availability of Key Experts and begin carrying out the Services not later than the number of days after the Effective Date specified in the SCC. |
| 14. Expiration of Contract | 14.1. | Unless terminated earlier pursuant to Clause GCC 19 hereof, this Contract shall expire at the end of such time period after the Effective Date as specified in the SCC. |
| 15. Entire Agreement | 15.1. | This Contract contains all covenants, stipulations and provisions agreed by the Parties. No agent or representative of either Party has authority to make, and the Parties shall not be bound by or be liable for, any statement, representation, promise or agreement not set forth herein. |
| 16. Modifications or Variations | 16.1. | Any modification or variation of the terms and conditions of this Contract, including any modification or variation of the scope of the Services, may only be made by written agreement between the Parties. However, each Party shall give due consideration to any proposals for modification or variation made by the other Party. |
| | 16.2. | In cases of substantial modifications or variations, the prior written consent of the Bank is required. |

17. Force Majeure

a. Définition

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| 17.1 | For the purposes of this Contract, “Force Majeure” means an event which is beyond the reasonable control of a Party, is not foreseeable, is unavoidable, and makes a Party’s performance of its obligations hereunder impossible or so impractical as reasonably to be considered impossible under the circumstances, and subject to those requirements, includes, but is not limited to, war, riots, civil disorder, earthquake, fire, explosion, storm, flood or other adverse weather conditions, strikes, lockouts or other industrial action confiscation or any other action by Government agencies. |
| 17.2. | Force Majeure shall not include (i) any event which is caused by the negligence or intentional action of a Party or such Party’s Experts, Sub-consultants or agents or employees, nor (ii) any event which a diligent Party could reasonably have been expected to both take into account at the time of the conclusion of this Contract, and avoid or overcome in the carrying out of its obligations hereunder. |
| 17.3. | Force Majeure shall not include insufficiency of funds or failure to make any payment required hereunder. |

b. No Breach of Contract

17.4 The failure of a Party to fulfill any of its obligations hereunder shall not be considered to be a breach of, or default under, this Contract insofar as such inability arises from an event of Force Majeure, provided that the Party affected by such an event has taken all reasonable precautions, due care and reasonable alternative measures, all with the objective of carrying out the terms and conditions of this Contract.

c. Measures to be Taken

17.5 A Party affected by an event of Force Majeure shall continue to perform its obligations under the Contract as far as is reasonably practical, and shall take all reasonable measures to minimize the consequences of any event of Force Majeure.

17.6 A Party affected by an event of Force Majeure shall notify the other Party of such event as soon as possible, and in any case not later than fourteen (14) calendar days following the occurrence of such event, providing evidence of the nature and cause of such event, and shall similarly give written notice of the restoration of normal conditions as soon as possible.

17.7 Any period within which a Party shall, pursuant to this Contract, complete any action or task, shall be extended for a period equal to the time during which such Party was unable to perform such action as a result of Force Majeure.

17.8 During the period of their inability to perform the Services as a result of an event of Force Majeure, the Consultant, upon instructions by the Client, shall either:

- (a) demobilize, in which case the Consultant shall be reimbursed for additional costs they reasonably and necessarily incurred, and, if required by the Client, in reactivating the Services; or
- (b) continue with the Services to the extent reasonably possible, in which case the Consultant shall continue to be paid under the terms of this Contract and be reimbursed for additional costs reasonably and necessarily incurred.
- (c) In the case of disagreement between the Parties as to the existence or extent of Force Majeure, the matter shall be settled according to Clauses GCC 44 & 45.

18. Suspension

18.1. The Client may, by written notice of suspension to the Consultant, suspend all payments to the Consultant hereunder if the Consultant fails to perform any of its obligations under this Contract, including the carrying out of the Services, provided that such notice of suspension (i) shall specify the nature of the failure, and (ii) shall request the Consultant to remedy such failure within a period not exceeding thirty (30) calendar days after receipt by the Consultant of such notice of suspension.

19. Termination

19.1. This Contract may be terminated by either Party as per provisions set up below:

a. By the Client

19.1.1. The Client may terminate this Contract in case of the occurrence of any of the events specified in paragraphs (a) through (f) of this Clause. In such an occurrence the Client shall give at least thirty (30) calendar days' written notice of termination to the Consultant in case of the events referred to in (a) through (d); at least sixty (60) calendar days' written notice in case of the event referred to in (e); and at least five (5) calendar days' written notice in case of the event referred to in (f):

- (a) If the Consultant fails to remedy a failure in the performance of its obligations hereunder, as specified in a notice of suspension pursuant to Clause GCC 18;
- (b) If the Consultant becomes (or, if the Consultant consists of more than one entity, if any of its members becomes) insolvent or bankrupt or enter into any agreements with their creditors for relief of debt or take advantage of any law for the benefit of debtors or go into liquidation or receivership whether compulsory or voluntary;
- (c) If the Consultant fails to comply with any final decision reached as a result of arbitration proceedings pursuant to Clause GCC 45.1;
- (d) If, as the result of Force Majeure, the Consultant is unable to perform a material portion of the Services for a period of not less than sixty (60) calendar days;
- (e) If the Client, in its sole discretion and for any reason whatsoever, decides to terminate this Contract;
- (f) If the Consultant fails to confirm availability of Key Experts as required in Clause GCC 13.

19.1.2. Furthermore, if the Client determines that the Consultant has engaged in corrupt, fraudulent, collusive, coercive [*or obstructive*] practices, in competing for or in executing the Contract, then the Client may, after giving fourteen (14) calendar days written notice to the Consultant, terminate the Consultant's employment under the Contract.

b. By the Consultant

19.1.3. The Consultant may terminate this Contract, by not less than thirty (30) calendar days' written notice to the Client, in case of the occurrence of any of the events specified in paragraphs (a) through (d) of this Clause.

- (a) If the Client fails to pay any money due to the Consultant pursuant to this Contract and not subject to dispute pursuant to Clause GCC 45.1 within forty-five (45) calendar days after

receiving written notice from the Consultant that such payment is overdue.

- (b) If, as the result of Force Majeure, the Consultant is unable to perform a material portion of the Services for a period of not less than sixty (60) calendar days.
- (c) If the Client fails to comply with any final decision reached as a result of arbitration pursuant to Clause GCC 45.1.
- (d) If the Client is in material breach of its obligations pursuant to this Contract and has not remedied the same within forty-five (45) days (or such longer period as the Consultant may have subsequently approved in writing) following the receipt by the Client of the Consultant's notice specifying such breach.

c. Cessation of Rights and Obligations

19.1.4. Upon termination of this Contract pursuant to Clauses GCC 12 or GCC 19 hereof, or upon expiration of this Contract pursuant to Clause GCC 14, all rights and obligations of the Parties hereunder shall cease, except (i) such rights and obligations as may have accrued on the date of termination or expiration, (ii) the obligation of confidentiality set forth in Clause GCC 22, (iii) the Consultant's obligation to permit inspection, copying and auditing of their accounts and records set forth in Clause GCC 25, and (iv) any right which a Party may have under the Applicable Law.

d. Cessation of Services

19.1.5. Upon termination of this Contract by notice of either Party to the other pursuant to Clauses GCC 19a or GCC 19b, the Consultant shall, immediately upon dispatch or receipt of such notice, take all necessary steps to bring the Services to a close in a prompt and orderly manner and shall make every reasonable effort to keep expenditures for this purpose to a minimum. With respect to documents prepared by the Consultant and equipment and materials furnished by the Client, the Consultant shall proceed as provided, respectively, by Clauses GCC 27 or GCC 28.

e. Payment upon Termination

19.1.6. Upon termination of this Contract, the Client shall make the following payments to the Consultant:

- (a) payment for Services satisfactorily performed prior to the effective date of termination; and
- (b) in the case of termination pursuant to paragraphs (d) and (e) of Clause GCC 19.1.1, reimbursement of any reasonable cost incidental to the prompt and orderly termination of this Contract, including the cost of the return travel of the Experts.

C. OBLIGATIONS OF THE CONSULTANT

20. General

a. Standard of Performance

20.1 The Consultant shall perform the Services and carry out the Services with all due diligence, efficiency and economy, in accordance with generally accepted professional standards and practices, and shall observe sound management practices, and employ appropriate technology and safe and effective equipment, machinery, materials and methods. The Consultant shall always act, in respect of any matter relating to this Contract or to the Services, as a faithful adviser to the Client, and shall at all times support and safeguard the Client's legitimate interests in any dealings with the third parties.

20.2. The Consultant shall employ and provide such qualified and experienced Experts and Sub-consultants as are required to carry out the Services.

20.3. The Consultant may subcontract part of the Services to an extent and with such Key Experts and Sub-consultants as may be approved in advance by the Client. Notwithstanding such approval, the Consultant shall retain full responsibility for the Services.

b. Law Applicable to Services

20.4. The Consultant shall perform the Services in accordance with the Contract and the Applicable Law and shall take all practicable steps to ensure that any of its Experts and Sub-consultants, comply with the Applicable Law.

20.5. Throughout the execution of the Contract, the Consultant shall comply with the import of goods and services prohibitions in the Client's country when

- (a) as a matter of law or official regulations, the Borrower's/Beneficiary's country prohibits commercial relations with that country; or
- (b) by an act of compliance with a decision of the United Nations Security Council taken under Chapter VII of the Charter of the United Nations, the Borrower's Country prohibits any import of goods from that country or any payments to any country, person, or entity in that country.

20.6. The Client shall notify the Consultant in writing of relevant local customs, and the Consultant shall, after such notification, respect such customs.

21. Conflict of Interests

21.1. The Consultant shall hold the Client's interests paramount, without any consideration for future work, and strictly avoid conflict with other assignments or their own corporate interests.

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| <p>a. Consultant Not to Benefit from Commissions, Discounts, etc.</p> | <p>21.1.1 The payment of the Consultant pursuant to GCC F (Clauses GCC 38 through 42) shall constitute the Consultant's only payment in connection with this Contract and, subject to Clause GCC 21.1.3, the Consultant shall not accept for its own benefit any trade commission, discount or similar payment in connection with activities pursuant to this Contract or in the discharge of its obligations hereunder, and the Consultant shall use its best efforts to ensure that any Sub-consultants, as well as the Experts and agents of either of them, similarly shall not receive any such additional payment.</p> <p>21.1.2 Furthermore, if the Consultant, as part of the Services, has the responsibility of advising the Client on the procurement of goods, works or services, the Consultant shall comply with the Bank's Applicable Guidelines, and shall at all times exercise such responsibility in the best interest of the Client. Any discounts or commissions obtained by the Consultant in the exercise of such procurement responsibility shall be for the account of the Client.</p> |
| <p>b. Consultant and Affiliates Not to Engage in Certain Activities</p> | <p>21.1.3 The Consultant agrees that, during the term of this Contract and after its termination, the Consultant and any entity affiliated with the Consultant, as well as any Sub-consultants and any entity affiliated with such Sub-consultants, shall be disqualified from providing goods, works or non-consulting services resulting from or directly related to the Consultant's Services for the preparation or implementation of the project, unless otherwise indicated in the SCC.</p> |
| <p>c. Prohibition of Conflicting Activities</p> | <p>21.1.4 The Consultant shall not engage, and shall cause its Experts as well as its Sub-consultants not to engage, either directly or indirectly, in any business or professional activities that would conflict with the activities assigned to them under this Contract.</p> |
| <p>d. Strict Duty to Disclose Conflicting Activities</p> | <p>21.1.5 The Consultant has an obligation and shall ensure that its Experts and Sub-consultants shall have an obligation to disclose any situation of actual or potential conflict that impacts their capacity to serve the best interest of their Client, or that may reasonably be perceived as having this effect. Failure to disclose said situations may lead to the disqualification of the Consultant or the termination of its Contract.</p> |
| <p>22. Confidentiality</p> | <p>22.1 Except with the prior written consent of the Client, the Consultant and the Experts shall not at any time communicate to any person or entity any confidential information acquired in the course of the Services, nor shall the Consultant and the Experts make public the recommendations formulated in the course of, or as a result of, the Services.</p> |

23. Liability of the Consultant	23.1	Subject to additional provisions, if any, set forth in the SCC, the Consultant's liability under this Contract shall be as determined under the Applicable Law.
24. Insurance to be Taken out by the Consultant	24.1	The Consultant (i) shall take out and maintain, and shall cause any Sub-consultants to take out and maintain, at its (or the Sub-consultants', as the case may be) own cost but on terms and conditions approved by the Client, insurance against the risks, and for the coverage specified in the SCC, and (ii) at the Client's request, shall provide evidence to the Client showing that such insurance has been taken out and maintained and that the current premiums therefore have been paid. The Consultant shall ensure that such insurance is in place prior to commencing the Services as stated in Clause GCC 13.
25. Accounting, Inspection and Auditing	25.1	The Consultant shall keep, and shall make all reasonable efforts to cause its Sub-consultants to keep, accurate and systematic accounts and records in respect of the Services and in such form and detail as will clearly identify relevant time changes and costs.
	25.2	The Consultant shall permit and shall cause its Sub-consultants to permit, the Bank and/or persons appointed by the Bank to inspect the Site and/or all accounts and records relating to the performance of the Contract and the submission of the Proposal to provide the Services, and to have such accounts and records audited by auditors appointed by the Bank if requested by the Bank. The Consultant's attention is drawn to Clause GCC 10 which provides, inter alia, that acts intended to materially impede the exercise of the Bank's inspection and audit rights provided for under this Clause GCC25.2 constitute a prohibited practice subject to contract termination (as well as to a determination of ineligibility under the Bank's prevailing sanctions procedures.)
26. Reporting Obligations	26.1	The Consultant shall submit to the Client the reports and documents specified in Appendix A, in the form, in the numbers and within the time periods set forth in the said Appendix.
27. Proprietary Rights of the Client in Reports and Records	27.1	Unless otherwise indicated in the SCC, all reports and relevant data and information such as maps, diagrams, plans, databases, other documents and software, supporting records or material compiled or prepared by the Consultant for the Client in the course of the Services shall be confidential and become and remain the absolute property of the Client. The Consultant shall, not later than upon termination or expiration of this Contract, deliver all such documents to the Client, together with a detailed inventory thereof. The Consultant may retain a copy of such documents, data and/or software but shall not use the same for

purposes unrelated to this Contract without prior written approval of the Client.

- 27.2 If license agreements are necessary or appropriate between the Consultant and third parties for purposes of development of the plans, drawings, specifications, designs, databases, other documents and software, the Consultant shall obtain the Client's prior written approval to such agreements, and the Client shall be entitled at its discretion to require recovering the expenses related to the development of the program(s) concerned. Other restrictions about the future use of these documents and software, if any, shall be specified in the SCC.

28. Equipment, Vehicles and Materials

- 28.1 Equipment, vehicles and materials made available to the Consultant by the Client, or purchased by the Consultant wholly or partly with funds provided by the Client, shall be the property of the Client and shall be marked accordingly. Upon termination or expiration of this Contract, the Consultant shall make available to the Client an inventory of such equipment, vehicles and materials and shall dispose of such equipment, vehicles and materials in accordance with the Client's instructions. While in possession of such equipment, vehicles and materials, the Consultant, unless otherwise instructed by the Client in writing, shall insure them at the expense of the Client in an amount equal to their full replacement value.
- 28.2 Any equipment or materials brought by the Consultant or its Experts into the Client's country for the use either for the project or personal use shall remain the property of the Consultant or the Experts concerned, as applicable.

D. CONSULTANT'S EXPERTS AND SUB-CONSULTANTS

29. Description of Key Experts

- 29.1 The title, agreed job description, minimum qualification and estimated period of engagement to carry out the Services of each of the Consultant's Key Experts are described in Appendix B.

30. Replacement of Key Experts

- 30.1 Except as the Client may otherwise agree in writing, no changes shall be made in the Key Experts.
- 30.2 Notwithstanding the above, the substitution of Key Experts during Contract execution may be considered only based on the Consultant's written request and due to circumstances outside the reasonable control of the Consultant, including but not limited to death or medical incapacity. In such case, the Consultant shall forthwith provide as a replacement, a person of equivalent or better qualifications and experience, and at the same rate of remuneration.

31. Removal of Experts or Sub-consultants

- 31.1 If the Client finds that any of the Experts or Sub-consultant has committed serious misconduct or has been charged with having committed a criminal action, or shall the Client determine that Consultant's Expert or Sub-consultant have engaged in corrupt, fraudulent, collusive, coercive [*or obstructive*] practice while performing the Services, the Consultant shall, at the Client's written request, provide a replacement.
- 31.2 In the event that any of Key Experts, Non-Key Experts or Sub-consultants is found by the Client to be incompetent or incapable in discharging assigned duties, the Client, specifying the grounds therefore, may request the Consultant to provide a replacement.
- 31.3 Any replacement of the removed Experts or Sub-consultants shall possess better qualifications and experience and shall be acceptable to the Client.
- 31.4 The Consultant shall bear all costs arising out of or incidental to any removal and/or replacement of such Experts.

E. OBLIGATIONS OF THE CLIENT**32. Assistance and Exemptions**

- 32.1 Unless otherwise specified in the SCC, the Client shall use its best efforts to:
- (a) Assist the Consultant with obtaining work permits and such other documents as shall be necessary to enable the Consultant to perform the Services.
 - (b) Assist the Consultant with promptly obtaining, for the Experts and, if appropriate, their eligible dependents, all necessary entry and exit visas, residence permits, exchange permits and any other documents required for their stay in the Client's country while carrying out the Services under the Contract.
 - (c) Facilitate prompt clearance through customs of any property required for the Services and of the personal effects of the Experts and their eligible dependents.
 - (d) Issue to officials, agents and representatives of the Government all such instructions and information as may be necessary or appropriate for the prompt and effective implementation of the Services.
 - (e) Assist the Consultant and the Experts and any Sub-consultants employed by the Consultant for the Services with obtaining exemption from any requirement to register or obtain any permit to practice their profession or to establish themselves either individually or as a corporate entity in the

Client's country according to the applicable law in the Client's country.

- (f) Assist the Consultant, any Sub-consultants and the Experts of either of them with obtaining the privilege, pursuant to the applicable law in the Client's country, of bringing into the Client's country reasonable amounts of foreign currency for the purposes of the Services or for the personal use of the Experts and of withdrawing any such amounts as may be earned therein by the Experts in the execution of the Services.
- (g) Provide to the Consultant any such other assistance as may be specified in the SCC.

33. Access to Project Site

- 33.1 The Client warrants that the Consultant shall have, free of charge, unimpeded access to the project site in respect of which access is required for the performance of the Services. The Client will be responsible for any damage to the project site or any property thereon resulting from such access and will indemnify the Consultant and each of the experts in respect of liability for any such damage, unless such damage is caused by the willful default or negligence of the Consultant or any Sub-consultants or the Experts of either of them.

34. Change in the Applicable Law Related to Taxes and Duties

- 34.1 If, after the date of this Contract, there is any change in the applicable law in the Client's country with respect to taxes and duties which increases or decreases the cost incurred by the Consultant in performing the Services, then the remuneration and reimbursable expenses otherwise payable to the Consultant under this Contract shall be increased or decreased accordingly by agreement between the Parties hereto, and corresponding adjustments shall be made to the Contract price amount specified in Clause GCC 38.1

35. Services, Facilities and Property of the Client

- 35.1 The Client shall make available to the Consultant and the Experts, for the purposes of the Services and free of any charge, the services, facilities and property described in the Terms of Reference (Appendix A) at the times and in the manner specified in said Appendix A.

36. Counterpart Personnel

- 36.1 The Client shall make available to the Consultant free of charge such professional and support counterpart personnel, to be nominated by the Client with the Consultant's advice, if specified in Appendix A.
- 36.2 Professional and support counterpart personnel, excluding Client's liaison personnel, shall work under the exclusive direction of the Consultant. If any member of the counterpart personnel fails to perform adequately any work assigned to such member by the Consultant that is consistent with the position occupied by such member, the Consultant

may request the replacement of such member, and the Client shall not unreasonably refuse to act upon such request.

**37. Payment
Obligation**

- 37.1 In consideration of the Services performed by the Consultant under this Contract, the Client shall make such payments to the Consultant for the deliverables specified in Appendix A and in such manner as is provided by GCC F below.

F. PAYMENTS TO THE CONSULTANT

38. Contract Price

- 38.1 The Contract price is fixed and is set forth in the SCC. The Contract price breakdown is provided in Appendix C.
- 38.2 Any change to the Contract price specified in Clause 38.1 can be made only if the Parties have agreed to the revised scope of Services pursuant to Clause GCC 16 and have amended in writing the Terms of Reference in Appendix A.

39. Taxes and Duties

- 39.1 The Consultant, Sub-consultants and Experts are responsible for meeting any and all tax liabilities arising out of the Contract unless it is stated otherwise in the SCC.
- 39.2 As an exception to the above and as stated in the SCC, all local identifiable indirect taxes (itemized and finalized at Contract negotiations) are reimbursed to the Consultant or are paid by the Client on behalf of the Consultant.

**40. Currency of
Payment**

- 40.1 Any payment under this Contract shall be made in the currency(ies) of the Contract.

**41. Mode of Billing and
Payment**

- 41.1 The total payments under this Contract shall not exceed the Contract price set forth in Clause GCC 38.1.
- 41.2 The payments under this Contract shall be made in lump-sum installments against deliverables specified in Appendix A. The payments will be made according to the payment schedule stated in the SCC.
- 41.2.1 Advance payment: Unless otherwise indicated in the SCC, an advance payment shall be made against an advance payment bank guarantee acceptable to the Client in an amount (or amounts) and in a currency (or currencies) specified in the SCC. Such guarantee (i) is to remain effective until the advance payment has been fully set off, and (ii) is to be in the form set forth in Appendix D, or in such other form as the Client shall have approved in writing. The advance payments will be set off by the Client in equal portions against the lump-sum installments specified in the SCC until said advance payments have been fully set off.
- 41.2.2 The Lump-Sum Installment Payments. The Client shall pay the Consultant within sixty (60) days after the receipt by the

Client of the deliverable(s) and the cover invoice for the related lump-sum installment payment. The payment can be withheld if the Client does not approve the submitted deliverable(s) as satisfactory in which case the Client shall provide comments to the Consultant within the same sixty (60) days period. The Consultant shall thereupon promptly make any necessary corrections, and thereafter the foregoing process shall be repeated.

41.2.3 *The Final Payment:* The final payment under this Clause shall be made only after the final report has been submitted by the Consultant and approved as satisfactory by the Client. The Services shall then be deemed completed and finally accepted by the Client. The last lump-sum installment shall be deemed approved for payment by the Client within ninety (90) calendar days after receipt of the final report by the Client unless the Client, within such ninety (90) calendar day period, gives written notice to the Consultant specifying in detail deficiencies in the Services, the final report. The Consultant shall thereupon promptly make any necessary corrections, and thereafter the foregoing process shall be repeated. 41.2.4 All payments under this Contract shall be made to the accounts of the Consultant specified in the SCC.

41.2.4 With the exception of the final payment under 41.2.3 above, payments do not constitute acceptance of the whole Services nor relieve the Consultant of any obligations hereunder.

42. Interest on Delayed Payments

42.1 If the Client had delayed payments beyond fifteen (15) days after the due date stated in Clause GCC 41.2.2, interest shall be paid to the Consultant on any amount due by, not paid on, such due date for each day of delay at the annual rate stated in the SCC.

G. FAIRNESS AND GOOD FAITH

43. Good Faith

43.1 The Parties undertake to act in good faith with respect to each other's rights under this Contract and to adopt all reasonable measures to ensure the realization of the objectives of this Contract.

H. SETTLEMENT OF DISPUTES

44. Amicable Settlement

44.1 The Parties shall seek to resolve any dispute amicably by mutual consultation.

44.2 If either Party objects to any action or inaction of the other Party, the objecting Party may file a written Notice of Dispute to the other Party providing in detail the basis of the dispute. The Party receiving the Notice of Dispute will

consider it and respond in writing within fourteen (14) days after receipt. If that Party fails to respond within fourteen (14) days, or the dispute cannot be amicably settled within fourteen (14) days following the response of that Party, Clause GCC 45.1 shall apply.

- 45. Dispute Resolution** 45.1 Any dispute between the Parties arising under or related to this Contract that cannot be settled amicably may be referred to by either Party to the adjudication/arbitration in accordance with the provisions specified in the SCC.

II. General Conditions

ATTACHMENT 1: Bank Policy – Corrupt and Fraudulent Practices

It is the Bank's policy to require that Borrowers (including beneficiaries of Bank Financing), consultants and their agents (whether declared or not), sub-contractors, sub-consultants, service providers or suppliers and any personnel thereof, observe the highest standard of ethics during the selection and execution of Bank-financed contracts¹. In pursuance of this policy, the Bank:

(a) defines, for the purposes of this provision, the terms set forth below as follows:

- (i) "corrupt practice" is the offering, giving, receiving or soliciting, directly or indirectly, of anything of value to influence improperly the actions of another party²;
- (ii) "fraudulent practice" is any act or omission, including misrepresentation, that knowingly or recklessly misleads, or attempts to mislead, a party³ to obtain financial or other benefit or to avoid an obligation;
- (iii) "collusive practice" is an arrangement between two or more parties⁴ designed to achieve an improper purpose, including to influence improperly the actions of another party;
- (iv) "coercive practice" is impairing or harming, or threatening to impair or harm, directly or indirectly, any party or the property of the party to influence improperly the actions of a party⁵;
- (v) "obstructive practice" is
 - (aa) deliberately destroying, falsifying, altering, or concealing of evidence material to the investigation or making false statements to investigators in order to materially impede a Bank investigation into allegations of a corrupt, fraudulent, coercive, or collusive practice; and/or threatening, harassing or intimidating any party to prevent it from disclosing its knowledge of matters relevant to the investigation or from pursuing the investigation, or

¹ In this context, any action taken by a consultant or any of its personnel, or its agents, or its sub-consultants, sub-contractors, service providers, suppliers, and/or their employees, to influence the selection process or contract execution for undue advantage, is improper.

² For the purpose of this sub-paragraph, "another party" refers to a public official acting in relation to the selection process or contract execution. In this context, "public official" includes Bank staff and employees of other organizations taking or reviewing selection decisions

³ For the purpose of this sub-paragraph, "party" refers to a public official; the terms "benefit" and "obligation" relate to the selection process or contract execution, and the "act or omission" is intended to influence the selection process or contract execution.

⁴ For the purpose of this sub-paragraph, "parties" refers to participants in the procurement or selection process (including public officials) attempting either themselves, or through another person or entity not participating in the procurement or selection process, to simulate competition or to establish contract prices at artificial, non-competitive levels, or are privy to each other's bid prices or other conditions.

⁵ For the purpose of this sub-paragraph, "party" refers to a participant in the selection process or contract execution.

- (bb) acts intended to materially impede the exercise of the Bank's inspection and audit rights provided for under paragraph 1.22(e) below.
- (b) will reject a proposal for award if it determines that the consultant recommended for award, or any of its personnel, or its agents, or its sub-consultants, sub-contractors, service providers, suppliers and/or their employees, has, directly or indirectly, engaged in corrupt, fraudulent, collusive, coercive, or obstructive practices in competing for the contract in question;
- (c) will declare misprocurement and cancel the portion of the Financing allocated to a contract if it determines at any time that representatives of the Borrower or of a recipient of any part of the proceeds of the Financing were engaged in corrupt, fraudulent, collusive, coercive, or obstructive practices during the selection process or the implementation of the contract in question, without the Borrower having taken timely and appropriate action satisfactory to the Bank to address such practices when they occur, including by failing to inform the Bank in a timely manner at the time they knew of the practices;
- (d) will sanction a firm or an individual, at any time, in accordance with prevailing Bank's sanctions procedures⁶, including by publicly declaring such firm or individual ineligible either indefinitely or for a stated period of time, (i) to be awarded a Bank-financed contract; and (ii) to be a nominated⁷ as sub-consultant, supplier, or service provider of an otherwise eligible firm being awarded a Bank-financed contract; and
- (e) will require that a clause be inserted in the RFP and in contracts financed by the Bank, requiring consultants and their agents, personnel, sub-consultants, sub-contractors, service providers, or suppliers, to permit the Bank to inspect all accounts, records and other documents relating to the submission of proposals and contract performance and to have them audited by auditors appointed by the Bank.

⁶ A firm or an individual may be declared ineligible to be awarded a Bank-financed contract (i) upon completion of the Bank's sanctions proceedings as per its sanctions procedures, including, inter alia, cross-debarment as agreed with other International Financial Institutions, including Multilateral Development Banks, or otherwise decided by the Bank, and through the application of the Proposal for the Implementation of a Sanctions Process within the African Development Bank; and (ii) as a result of temporary suspension or early temporary suspension in connection with an on-going sanctions proceeding.

⁷ A nominated sub-consultant, supplier, or service provider is one which has been either: (i) included by the consultant in its proposal because it brings specific and critical experience and know-how that are accounted for in the technical evaluation of the consultant's proposal for the particular services, or (ii) appointed by the Borrower.

III. Special Conditions of Contract

Number of GC Clause	Amendments of, and Supplements to, Clauses in the General Conditions of Contract
1.1(b) and 3.1	The Contract shall be construed in accordance with the law of the United Kingdom
4.1	The language is: <i>English.</i>
6.1 and 6.2	The addresses are: Client: Serious Shea SARL Attention: Mr. Alastair Woods Facsimile: _____ E-mail (where permitted): alastair.woods@impactagri.com Consultant: Sensonomic AS Attention: Anders Gundersen Facsimile: _____ E-mail: anders.gundersen@sensonomic.com
8.1	The Lead Member on behalf of the JV is <i>N/A</i>
9.1	The Authorized Representatives are: For the Client: Mr. Alastair Woods, Lead Coordinator Serious Shea For the Consultant: Mr. Anders Gundersen
11.1	The effectiveness conditions are the following: <i>"N/A"</i>
12.1	Termination of Contract for Failure to Become Effective: The time period shall be 20 days
13.1	Commencement of Services: The number of days shall be within 7 days Confirmation of Key Experts' availability to start the Assignment shall be submitted to the Client in writing as a written statement signed by each Key Expert.

14.1	Expiration of Contract: The time period shall be 12 <i>months</i>
21 b.	The Client reserves the right to determine on a case-by-case basis whether the Consultant should be disqualified from providing goods, works or non-consulting services due to a conflict of a nature described in Clause GCC 21.1.3 Yes
23.1	No additional provisions. [OR The following limitation of the Consultant's Liability towards the Client can be subject to the Contract's negotiations: “Limitation of the Consultant's Liability towards the Client: (a) Except in the case of gross negligence or willful misconduct on the part of the Consultant or on the part of any person or a firm acting on behalf of the Consultant in carrying out the Services, the Consultant, with respect to damage caused by the Consultant to the Client's property, shall not be liable to the Client: (i) for any indirect or consequential loss or damage; and (ii) for any direct loss or damage that exceeds 1 times the total value of the Contract; (b) This limitation of liability shall not (i) affect the Consultant's liability, if any, for damage to Third Parties caused by the Consultant or any person or firm acting on behalf of the Consultant in carrying out the Services; (ii) be construed as providing the Consultant with any limitation or exclusion from liability which is prohibited by United Kingdom law
24.1	The insurance coverage against the risks shall be as follows: (a) Professional liability insurance, with a minimum coverage of not less than US\$ 120,000

27.2	The Consultant shall not use <i>any material (both input and output)</i> for purposes unrelated to this Contract without the prior written approval of the Client.
38.1	The Contract price is: US\$ 149,750 inclusive of local indirect taxes. Any indirect local taxes chargeable in respect of this Contract for the Services provided by the Consultant shall “reimbursed” by the Client “to” the Consultant.
39.1 and 39.2	The Client warrants that <i>“the Client shall reimburse the Consultant, the Sub-consultants and the Experts”</i> any indirect taxes, duties, fees, levies and other impositions imposed, under the applicable law in the Client’s country, on the Consultant, the Sub-consultants and the Experts in respect of: (a) any payments whatsoever made to the Consultant, Sub-consultants and the Experts (other than nationals or permanent residents of the Client’s country), in connection with the carrying out of the Services; (b) any equipment, materials and supplies brought into the Client’s country by the Consultant or Sub-consultants for the purpose of carrying out the Services and which, after having been brought into such territories, will be subsequently withdrawn by them; (c) any equipment imported for the purpose of carrying out the Services and paid for out of funds provided by the Client and which is treated as property of the Client; (d) any property brought into the Client’s country by the Consultant, any Sub-consultants or the Experts (other than nationals or permanent residents of the Client’s country), or the eligible dependents of such experts for their personal use and which will subsequently be withdrawn by them upon their respective departure from the Client’s country, provided that: i) the Consultant, Sub-consultants and experts shall follow the usual customs procedures of the Client’s country in importing property into the Client’s country; and ii) if the Consultant, Sub-consultants or Experts do not withdraw but dispose of any property in the Client’s country upon which customs duties and taxes have been exempted,

	the Consultant, Sub-consultants or Experts, as the case may be, (a) shall bear such customs duties and taxes in conformity with the regulations of the Client's country, or (b) shall reimburse them to the Client if they were paid by the Client at the time the property in question was brought into the Client's country.
41.2	The payment schedule: ● 1st Payment: Delivery of work plan report and team mobilisation: US\$29,950 (20% of contract) ● 2nd Payment: Completion of Task 3.1 (D1, D2, and D3): Classifying zones and preparation of GIS platform US\$52,412.50 (35% of contract) ● 3rd Payment: completion of Task 3.2 and 3.3 (D5, D6, D7, D8, D10 and D12): Field Data collection: validating and characterising zones: US\$22,462.50 (15% of contract) ● 4th Payment: Completion of Task 3.4 Shea Management plan (D13 and D14): US\$22,462.50 (15% of contract) ● 5th Payment: Finalise data and software for integration with other outputs (D11): US\$22,462.50 (15% of contract)
41.2.4	The accounts are: Account name: Sensonomic AS Account address: Ole Vigs gate 5, 5006 Bergen, Norway IBAN: NO0615035372897 SWIFT: DNBANOKKXXX Bank name & address: DNB, Postboks 1600 Sentrum, 0021 Oslo
42.1	The interest rate is: N/A
45.1	Note: In contracts with foreign consultants, the Bank requires that the international commercial arbitration in a neutral venue is used. Disputes shall be settled by arbitration in accordance with the following provisions: 1. <u>Selection of Arbitrators.</u> Each dispute submitted by a Party to arbitration shall be heard by a sole arbitrator or an arbitration panel composed of three (3) arbitrators, in accordance with the following provisions:

	(a) Where the Parties agree that the dispute concerns a technical matter, they may agree to appoint a sole arbitrator or, failing agreement on the identity of such sole arbitrator within thirty (30) days after receipt by the other Party of the proposal of a name for such an appointment by the Party who initiated the proceedings, either Party may apply to <i>the Federation Internationale des Ingenieurs-Conseil (FIDIC) of Lausanne, Switzerland</i> for a list of not fewer than five (5) nominees and, on receipt of such list, the Parties shall alternately strike names therefrom, and the last remaining nominee on the list shall be the sole arbitrator for the matter in dispute. If the last remaining nominee has not been determined in this manner within sixty (60) days of the date of the list, <i>the Federation Internationale des Ingenieurs-Conseil (FIDIC) of Lausanne, Switzerland</i> shall appoint, upon the request of either Party and from such list or otherwise, a sole arbitrator for the matter in dispute. (b) Where the Parties do not agree that the dispute concerns a technical matter, the Client and the Consultant shall each appoint one (1) arbitrator, and these two arbitrators shall jointly appoint a third arbitrator, who shall chair the arbitration panel. If the arbitrators named by the Parties do not succeed in appointing a third arbitrator within thirty (30) days after the latter of the two (2) arbitrators named by the Parties has been appointed, the third arbitrator shall, at the request of either Party, be appointed by <i>the Secretary General of the Permanent Court of Arbitration, The Hague</i> (c) If, in a dispute subject to paragraph (b) above, one Party fails to appoint its arbitrator within thirty (30) days after the other Party has appointed its arbitrator, the Party which has named an arbitrator may apply to <i>the Secretary General of the Permanent Court of Arbitration, The Hague</i> to appoint a sole arbitrator for the matter in dispute, and the arbitrator appointed pursuant to such application shall be the sole arbitrator for that dispute.
	2. <u>Rules of Procedure.</u> Except as otherwise stated herein, arbitration proceedings shall be conducted in accordance with the rules of procedure for arbitration of the United Nations Commission on International Trade Law (UNCITRAL) as in force on the date of this Contract. 3. <u>Substitute Arbitrators.</u> If for any reason an arbitrator is unable to perform his/her function, a substitute shall be appointed in the same manner as the original arbitrator. 4. <u>Nationality and Qualifications of Arbitrators.</u> The sole arbitrator or the third arbitrator appointed pursuant to paragraphs 1(a) through 1(c) above shall be an internationally recognized legal or technical expert

	with extensive experience in relation to the matter in dispute and shall not be a national of the Consultant's home country or of the Government's country. For the purposes of this Clause, "home country" means any of: (a) the country of incorporation of the Consultant or of any of their members or Parties; or (b) the country in which the Consultant's or any of their members' or Parties' principal place of business is located; or (c) the country of nationality of a majority of the Consultant's or of any members' or Parties' shareholders; or (d) the country of nationality of the Sub-consultants concerned, where the dispute involves a subcontract.
	5. <u>Miscellaneous.</u> In any arbitration proceeding hereunder: (a) proceedings shall, unless otherwise agreed by the Parties, be held in <i>the Secretary General of the Permanent Court of Arbitration, The Hague</i> (b) the <i>English</i> language shall be the official language for all purposes; and (c) the decision of the sole arbitrator or of a majority of the arbitrators (or of the third arbitrator if there is no such majority) shall be final and binding and shall be enforceable in any court of competent jurisdiction, and the Parties hereby waive any objections to or claims of immunity in respect of such enforcement.

APPENDICES

APPENDIX A – TERMS OF REFERENCE

1.1. Background

Shea processing is a major industry in Burkina Faso, and conducted almost exclusively by women's groups; however, it still involves outdated methods: potentially degrading habitats including shea trees, emitting significant greenhouse gas emissions, causing a number of health and hygiene risks, and not realising its potential to support social and -economic development.

A consortium including an international project developer, a renowned local shea processing training company, and an engineering firm have established **Serious Shea**, which seeks to improve the shea butter-making process in Burkina Faso.

The project has received the firm support of the Ministry of Environment in Burkina Faso, and grant funding from the African Development Bank and World Bank to conduct the preparation activities, and will ensure women's groups can produce the shea butter more efficiently and sustainably, and also achieve significantly better access-to-market (receiving 2x - 5x more profit per tonne). The project will use clean technology at each processing centre and drive reforestation / afforestation at the collection sites, generating certified carbon credits by decarbonising the process and REDD+.

1.2. Design

Serious Shea has been established to enable the sustainable sourcing, processing and export of shea butter, displacing traditional methods, which drive significant deforestation and greenhouse gas emissions, and offer no socio-economic improvement to the women's groups.

The foundation of the agribusiness is the implementation of clean and efficient processing centres, known as *Eco-Processing Centres* where the training and technology is provided to the women's groups to collect the shea nuts sustainably, and process shea butter cleanly and efficiently.

The project will transform the shea sector in Burkina Faso, and enable the production and export of high volume and high quality sustainable shea butter, where the product is facing rapidly increasing demand from cosmetic and food (e.g. chocolate) markets. Importantly, local women's groups are core partners in Serious Shea, and will achieve significantly better access-to-market and 2x-5x higher income.

1.3. Preparation Activities

Serious Shea has secured grant support from the AFT Fund (Agricultural Fast-Track Fund), established by the AfDB (African Development Bank), which will fund key preparation activities for the agribusiness. Serious Shea therefore has grant funding support to complete preparatory planning and assessment, including business planning, forestry mapping, site feasibility assessments, and the technology designs. The key objective in completing this work is to assist Serious Shea in raising funds for implementation.

More specifically, the preparation work includes the following:

Component 1: A Validated and Bankable Business Plan

- Validate key financial assumptions, including capital costs, production volumes, operational costs, etc.
- Engage potential off-takers to estimate purchasing requirements, prices, etc. for financial model
- Develop and finalise financial model, including cash flow, P&L, balance sheet and potential optimal funding structure
- Finalise draft business plan, outlining core strategy, key activities, logistics, team, risks management, financial projections, investment requirements, etc.

Component 2: Site Feasibility Assessments

- Identify preferred 60 collection sites and location of 10 processing centres based on assessment of the concentration of women's groups, shea production areas, REDD+ regions, existing transportation networks, processing volumes, social assessment of communities and women's groups surrounding each collecting site
- Complete optimal design for logistics of Serious Shea's supply chain, which includes the process of collecting the fruit/nuts, storing the fruit/nuts, legal agreements with suppliers, payment processes, processed butter logistics, and export logistics

Component 3: Tree Mapping and Forestry Assessments

- Identify and classify the best areas for collection and processing of shea fruits
- Characterise each collecting area based on habitat and potential harvest. This objective is to develop a baseline of the existing shea habitat and associated drivers that are influencing production (e.g. fire, climate, agricultural practices). The objective is to also identify maximum, minimum and most likely volumes of shea nuts that could be sourced from each of the 60 collection areas
- Develop shea management plan to be implemented at each collecting centre to ensure sustainable shea harvesting, inform any tree or habitat improvement. This includes ways to influence mentalities/taboo about shea trees, training on grafting/pruning/etc, collective action for best shea management practices (collection dates, fire control, livestock encroachment, etc), maybe the set up of women's shea plots with access to land "ownership" through village-level conventions and endorsement from traditional village authorities, etc. This will draw from the vegetation baseline (Component 3) and social baseline (Component 2)
- Establish a Geographical Information System (GIS) as a management tool to track shea production, harvesting and impact of activities

Component 4: Site Technology Design and Development

- Outline potential technological options for collecting, storing, processing and transporting of shea butter on a large-scale across Burkina Faso
- Refine potential solutions with key stakeholders (including Serious Shea and the women's groups), rank/prioritise preferred solutions and agree preferred technical design
- Prepare detailed final designs, costings, and implementation plan for 60 collection sites, storage facility, transport logistics and 10 processing centres

This document refers exclusively to **Component 3: Tree Mapping and Forestry Assessment**, which is covered in more detail in the following sections.

2. Goals & Objectives of Serious Shea Tree Mapping and Forestry Assessment

Component 3 of the AFT grant focuses on tree mapping and forestry assessment. The overarching goals are to:

- Validate the optimal areas for the sustainable collection and transportation of shea fruits for processing based on field vegetation survey
- Characterise each collecting area based on habitat and potential harvest. This objective is to develop a baseline of the existing shea habitat and associated drivers that are influencing production (e.g. fire, climate, agricultural practices). The objective is to also identify maximum, minimum and most likely volumes of shea nuts that could be sourced from each of the 60 collection areas
- Develop shea management plan to be implemented at each collecting centre to ensure sustainable shea harvesting, inform any tree or habitat improvement. This includes ways to influence mentalities/taboos about shea trees, training on grafting/pruning/etc, collective action for best shea management practices (collection dates, fire control, livestock encroachment, etc), maybe the set up of women's shea plots with access to land "ownership" through village-level conventions and endorsement from traditional village authorities, etc. This will draw from the vegetation baseline (Component 3) and social baseline (Component 2)
- Integrate results into Geographical Information System (GIS) as a management tool to track shea production, harvesting and impact of activities

3. Scope of Services, Tasks, and Expected Deliverables

3.1. Broad-Scale Classification of Habitats Associated with Collection Sites and Preparation of GIS Platform

3.1.1. Activities, Components and Tasks

- Engage Ministry of Environment (MoE) to understand the agroforestry efforts in Burkina Faso to date, and collaborate fully on the tree mapping and forestry assessment. The MoE is working on shea tree regeneration and planting, and has conducted multiple studies, so will be able to share agronomic, yield, and production cycle data.
- Compile a series of spatially explicit data layers that can inform potential shea production in Burkina Faso and profile various areas that may form the basis for classifying Serious Shea's preferred 60 collection sites
- Identify the priority areas for the development of forestry program by the Burkina Faso government and international donors (PIF/REDD+)
- Classify each of the 60 potential collection-sites in terms of bioclimatic zones. This will use a brief literature review to identify spatial data related to variables such as soils, precipitation, fire, land-use, drainage, temperature, elevation
- Prepare land-use/land cover maps of areas surrounding each of the 60 collection site using remote sensing. The classifications will be derived from participatory mapping activities but could include young (<5 years), medium-aged (5-9 years), old fields (>10 years including either village or bush fields); burned/unburned woodlands and natural forest
- The classifications will be ground truthed through use of data collected in field activities. Inform the team conducting field work the preferred location for conducting transects. Work with the remote sensing team to explore correlations between spatial data strata and locally recognized land use classes in the participatory mapping exercise to locate most relevant transects and sample plots accordingly
- The results from the remote sensing team will be presented in both raster and shape files within an accessible GIS for Serious Shea

3.1.2. Proposed Method of Execution

- Organize a kick-off meeting with the Serious Shea management team and Ministry of Environment in Ouagadougou
- Organize a weekly update with a representative of the Serious Shea management team
- Provide a monthly report on progress
- The initial stratification will field proofing to validate the assumption on the GIS map of the most productive areas/ Women cooperative density / REDD+ selected areas
- The secondary stratification will use existing habitat/land cover maps and update those with contemporary remote sensing data

- The task will require interaction with the field team to validate the stratification through their ground-truthing activities. This can include extracting high definition images of each transects to be used in the field
- The Remote Sensing team should be able to provide interpretable” satellite images prior to the field work to question villagers about land use / management practices in each strata shown on the image

3.1.3. Expected Deliverables

- GIS collection of thematic spatial layers informing potential shea production and REDD+ (D1)
- Preparation of land-use/land cover, and bioclimatic maps surrounding each of the 60 collection sites based on secondary data and remote sensing data (D2)
- Method statement to allow replication of analysis (D3)
- This task will need to be conducted in collaboration with national forestry institutions and the Ministry of Environment as identified by Serious Shea partners

3.1.4. Capacity Building

- The GIS system will be used by Serious Shea partners and the Burkina Faso-based team as a decision support system.

3.2. Validating and Characterising Shea Tree Populations within Collecting Areas

3.2.1. Activities, Components and Tasks

- Prepare high-resolution images using remote sensing for each collection site for use by the field team
- As part of the preparation work, undertake participatory mapping within 3 collection sites (identified through Component 2), including:
 - Exploring the general land and tree use / management practices
 - Assessing how each women’s group serves the collection site (i.e. demarcate and georeference the areas where they collect)
- Development of vegetation survey methodology. This will be based on transects covering meaningful spatial land management units with plots laid out on either side at distance intervals to be determined locally with plot size varying according to vegetation type/density (larger field plots compared to fallow plots). Tree parameters measured in plots include: a) species; b) tree diameter at breast height (DBH), total height); c) existing land use; latitude and longitude readings by GPS; d) Geo-referenced photos of each shea tree; f) marks of tree pruning; g) evidence of fire
- The vegetation survey could include 3-4 transects per 12 collection sites with 30-50 transects conducted in total.
- Conduct semi-structured interviews with each farmer “owner” of the field / fallow / bush area surveyed above including history of cultivation and fallow, tenure, land and tree management practices, household variables, etc
- Integrate results into project GIS system

3.2.2. Proposed Method of Execution

- Organize a kick-off meeting with the Serious Shea management team in Ouagadougou
- Organize a weekly update with a representative of the Serious Shea management team
- Provide a monthly report on progress
- This task needs to align with the social assessment which will also be conducting a participatory appraisal with women's groups and other interviews. This includes aligning the timing and integrating survey instrument.
- The findings need to be integrated into the GIS platform and decision-support software developed as part of Component 2, allowing for ongoing assessment and operational decision-support
- Each vegetation survey team will need to include in costing dbh tapes, clinometers, GPS. They will also need to use a strong stake with a string attached to easily determine circular plots for fallows (short string) and fields (long string). Each team should include a technician with local botanical knowledge.
- This task will need to be conducted in collaboration with national forestry institutions and the Ministry of Environment as identified by Serious Shea partners

3.2.3. *Expected Deliverables*

- RS data covering each of the 57 remaining collection sites (D-5)
- Validation on 3 collection sites for adjustments in object recognition in RS data (D-6)
- Detailed assessment of village-level shea-based tree populations and key drivers affecting shea management (D-7)
- GIS maps of demarcated zones for each collection area and areas of operation for women's groups likely to use each collection site (D-8)

3.2.4. *Capacity Building*

- Field-based work includes the foundation for ongoing monitoring and evaluation of shea collection overseen by Serious Shea

3.3. Assessment of Potential Yield and Nut Quality

3.3.1. *Activities, Components and Tasks*

- Develop and draft methodology to monitor shea fruit yield of individual trees based on selection criteria (local preference, perceived productivity, location, etc). A maximum number of monitored trees will include 10-15 trees in each of up to six land use categories within each of up to six agroclimatic zones. (15 x 6 x 6 = 540 trees in total). The methodology will involve individual women recording volumes collected from individual trees.
- Organize training sessions with school-going educated village youth from all target locations at central locations to carry out the monitoring methodology and

support women during the growing season to apply it and check the validity and document results.

- Pilot approach with a specific women's group and develop approach to engage with other collecting sites
- Support the implementation of data and sample collection by 15 women's groups during the 2018 harvest period
- Assess the potential minimum, maximum and most likely yields for each collection site based on information above.
- Draft approach to train individuals from 60 women's groups at a central location to conduct and record tree yields. The approach will likely be implemented in partnership with national forestry institutions who will need to have an input into this process. The team and project partners will need to support and monitor them for the duration of the project. This approach has to be designed such that it can be extended by other national collaborators to the AFT project
- For samples collected, assess fat content, sterin content and characteristic of nut. For trees for which we will assess the chemical composition of nut/butter, the field team will also collect data on additional variables including crown shape, height to 1st branch, leaf characteristics (incl. petiole), fruit characteristics, nut characteristics.

3.3.2. *Proposed Method of Execution*

- The trees should be uniquely identified to enable future selection and propagation activities
- Yield data could be collected by women's groups
- Yield data for zones should also be aligned with information collected through Ecodata survey (conducted May 2018) which asks women's groups to document total volume collected and historical production
- This task will need to be conducted in collaboration with national forestry institutions and the Ministry of Environment as identified by Serious Shea partners

3.3.3. *Expected Deliverables*

- Assessment of annual fruit yields (D-10)
- Training approach for assessing yield across multiple collection sites
- Integrate findings with GIS platform and decision support platform (D-12)

3.3.4. *Capacity Building*

- This task closely links with existing collector's activities to promote the best trees. It will provide a framework for capturing existing informal knowledge.

3.4. Develop Shea Management Plan and REDD+ Opportunities

3.4.1. *Activities, Components and Tasks*

- GIS mapping of priority collection areas located in REDD+ regions
- Develop shea forest management plan. This is conducted by third party forester and includes interventions to be considered for parkland, farmland etc. This will include supporting natural regeneration (grafting, crown/pruning), opportunities for planting. It also includes associated environmental (and carbon benefit for implementing plan.
- Identify measures that could be replicated across separate collections sites (and associated ‘zones’).
- Develop approach to ensure ongoing learning and adaptive management at the level of each collection site.
- Engage with and obtain feedback from key stakeholders identified by Serious Shea partners into the draft approach and prepare final document.

3.4.2. *Proposed Method of Execution*

- Organize a kick-off meeting with the Serious Shea management team in Ouagadougou
- Organize a weekly update with a representative of the Serious Shea management team
- Provide a monthly report on progress
- This will be built on existing published materials and results from engagement with women’s groups through the AFT grant
- This task will need to be conducted in collaboration with national forestry institutions and the Ministry of Environment as identified by Serious Shea partners

3.4.3. *Expected Deliverables*

-
- Develop Shea Management Plan and REDD+ Opportunities (D-13 and D-14). The draft management plan includes a 10-20 page shea management plan (in English and French) and an annex including a series of templates for specific actions (e.g. nature of action, how it is conducted (including timing), when it should be considered, training/capacity building required, equipment required) with separate shorter practical versions for women's groups and company management
-

3.4.4. *Capacity Building*

- This deliverable will be used as an ongoing tool to support decision-making and management.

Team Composition and Qualification Requirements for the Key Experts

The team needs to consist of:

- Mapping, remote sensing and GIS specialists (>10 years' experience in remote sensing). This should include an ability to integrate into the decision-support software developed in Component 2
- Field assessment team with track record of conducting primary research in West Africa (>5 years' experience in forestry research, with a degree in forestry or ecology). This should also include field technicians familiar with conducting vegetation surveys
- Shea management / forestry expert (>5 years' experience in the shea sector in West Africa). This expert should have an established international track record in understanding key drivers underlying shea production. They should be able to design vegetation and social surveys. The person should be able to train field assessment teams, and have experience in processing results and preparing outputs to support Serious Shea decision making.

4. Reporting Requirements and Time Schedule for Deliverables

The key deliverables will need to be delivered:

- **Task 3.1:** Classifying zones and preparation of GIS platform (18th March 2019)
- **Task 3.2:** Field Data collection (1st July 2019)
- **Task 3.3:** Shea Management plan and Final Report (9th September 2019)

All results need to also include a summary format due on 19th November 2019

APPENDIX B – KEY EXPERTS

*CVs of key experts are included at the end of the contract.

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APPENDIX C – BREAKDOWN OF CONTRACT PRICE

We hereby agree that we have agreed to pay to the Experts listed, who will be involved in performing the Services. The reimbursable expenses are also included below:

No.	Name	Position (as in TECH-6)	Person-month Remuneration Rate (US Dollar)	Time Input in Person/Days (from TECH-6)	TOTAL (US Dollars)
—	Key Experts				
K-1	Lasse Gullvåg Sætre	Project Lead	[Home] 650/day	84	54,600
				0	0
K-2	Jean-Marc Boffa	Lead agroforestry expert	[Home] 1,000/day	15	15,000
			[Field] 1,000/day	23	23,000
K-3	Joanna Wolstenholme	RS analyst	[Home] 650/day	20	13,000
				0	0
K-4	Adam Formica	Data scientist	[Home] 650/day	11	7,150
				0	0
—	Non-Key Experts				
N-1	Brigitte Bastide	Local agroforestry supervisor	[Home] 207.61/day	4	830
			[Field] 207.61/day	20	4152
N-2	Four forestry BSc graduates	Forestry enumeration	346.02/month	160	2768
				0	0
N-3	Forestry graduates	Fruit yield monitoring	86.51	36	3,114
				0	0
N-4	Field Daily Labor	Village field assistants	3.46/day	200	692
				0	0
N-5	Labor for monitoring tree yields	Village field assistants	51.90/month	36	1,869
				0	0
	Total Costs (US \$)				126,175

B. [Reimbursable]					
N°	Type of [Reimbursable Expenses]	Unit	Unit Cost (USD)	Quantity	Total (USD)
	Per diem allowances Jean-Marc Boffa - Ouagadougou	Day	144.00	16	2,304
	Per diem allowances Jean-Marc Boffa - elsewhere	Day	87.00	11	957
	Per diem allowances Brigitte Bastide	Day	87.00	20	1,740
	Forest enumerators	Month	432.53	8	3,460
	Fruit production monitors	Day	30.00	36	1,080
	International flights - Jean-Marc Boffa - MLA-OUA	Trip	1700.00	3	5,100
	4WD vehicle rental	Day	86.51	44	3,806
	Vehicle fuel, 10.000cfa/100 km	Cost of fuel/100 km	17.30	50	865
	Motorcycle rental, forest enumeration 5,000cfa/day	Month	190.31	4	761
	Motorcycle rental, fruit production, 5,000cfa/day	Day	8.65	33	285
	Motorcycle fuel for forestry enumeration	Cost of fuel/100km	10.22	33	337
	Motorcycle fuel for fruit production monitoring	Cost of fuel/100km	10.22	120	1,227
	Equipment (2 GPS, 2 dbh tapes, 2 smartphones for data entry, 3 x 50m drag tapes, clipboards, backpacks)	Items	890	1	890
	Communication, telephone, internet, bank fees	Various	610	1	610
	Contingency	Various	151	1	151
Total Costs					23,575

The total Contract Cost is **US\$149,750**.

Signature: 

Name and title: Jon White, Director of Serious Shea SARL

Date: 23/12/2018

sensonomic

Lasse Gullvåg Sætre

Before finding his home in Sensonomic, Lasse worked as GIS analyst and cartographer for the Norwegian Railway Directorate. He has worked in the software sector for many years, after joining the free software movement in his teens. He is currently on a member of the IFAD consortium for Sensonomic.

Education and Qualifications

2017	University of Oslo TIK Centre for Technology, Innovation and Culture, Oslo	Master of Science and Technology Studies (STS)
2015	University of Oslo Department of Sociology and Human Geography	Bachelor of Human Geography

Experience

2018 -	Sensonomic AS <i>Geographer</i>
2018	University of Oslo <i>Research Officer</i>
2017	Norwegian Railway Directorate <i>GIS analyst and Cartographer</i>
2014 - 2017	University of Oslo <i>GIS seminar leader and lecturer</i>
2012 - 2014	Kilden genderresearch.no <i>Science journalist</i>
2006 - 2012	Get AS <i>Business Analyst</i>

Selected Projects

2018 - 2019 **International Fund for Agricultural Development, United Nations**

Climate change adaptation for rural smallholders

Sensonomic is project lead on a three year project projected to positively impact the lives of 30,000 people in Mali, Senegal, and Cameroon. The project is sponsored through a large grant from UN-IFAD, and interlinks with other government supported projects in the target countries.

2018 **University of Oslo**

Fishery Industries and Sciences

Worked as organizer and did daily administration of the research project "ERC StG: Enacting the good economy: Biocapitalization and the little tools of valuation", where cod farming is studied through the lense of valuation practices of the 'blue revolution' in late-capitalism.

2017 **Greenpeace Norway**

People vs. Arctic oil climate lawsuit

GIS consultant mapping the Arctic sea ice, its extent, withdrawal, and the Norwegian petroleum infrastructure. The resulting maps were used as evidence in the high-profile lawsuit against the Norwegian government.

2017 **Norwegian Railway Directorate**

The Oslo Concept Study (KVU)

Around 10 billion dollars is budgeted for upgrading the Oslo transportation network in the coming 15 to 45 years. Lasse mapped commute trends, population growth and business developments with regards to new train stations in Oslo.

Trivia

- Regular speaker at the Free and Open Source Software for Geoinformatics, Norway
- Organised the collaborative mapping initiative Maptime Oslo since 2015
- Novice sailor

Adam Formica

Adam's work on field ecology experiments during his undergraduate years at Columbia University opened his eyes to the many ways of quantifying and monitoring environmental data. He went on to pursue a Master's in Biodiversity Conservation at the University of Oxford to learn how to put these data to use in protecting species and landscapes. He is now continuing at Oxford with a PhD in Geography exploring how computational simulation can help to increase agricultural efficiency and reduce deforestation. In between his studies, he worked at Microsoft Research and the University of East Anglia. He is excited by the potential of new data sources and technologies to create a world better for nature and people.

Education and Qualifications

2016 -	University of Oxford Department of Geography, Oxford, UK	D.Phil., Geography & the Environment
2013 - 2015	University of Oxford Department of Geography, Oxford, UK	M.Phil., Geography & the Environment, <i>Distinction</i>
2009 - 2013	Columbia University United States	BA, Environmental Science, Magna cum <i>laude</i>

Experience

2017 -	Sensonomic Limited <i>Senior Modeller, Build computational models to better manage land assets.</i>
2016	University of East Anglia, Norwich, UK <i>Research Officer - 'Sustainable management of the Asian Houbara'</i>
2014	Microsoft Research Cambridge, UK <i>Research Officer - 'Environmental and social trade-offs for REDD+ prioritization'</i>

Selected Projects

- 2018 - International Partnership Programme, UK Space Agency**
CommonSensing - remote sensing for climate resilience
Sensonomic is thematic lead for agritech on a three year project for the small independent island states of Fiji, Vanuatu and Solomon Islands. The goal is to develop a service for resilient agriculture based on remote sensing and computational simulations. Project partners include the Commonwealth Secretariat, UNOSAT, SA Catapult and UK Met Office.
- 2016 - 2019 International Fund for Agricultural Development, United Nations**
Climate change adaptation for rural smallholders
Sensonomic is project lead on a three year project projected to positively impact the lives of 30,000 people in Mali, Senegal, and Cameroon. The project is sponsored through a large grant from UN-IFAD, and interlinks with other government supported projects in the target countries.
- Summer 2013 Environmental Defense Fund, San Francisco, CA**
Reviewed wetlands and clean energy literature for Chief Scientist's office
- Summer 2012 University of Alaska, Fairbanks, AK**
'Quantifying the physiology of structurally complex arctic vegetation and implications for carbon cycling in a shrubbier tundra'
- Summer 2011 University of Colorado, Boulder, CO**
'Mapping woody encroachment on the Niwot Ridge saddle'
- Trivia**
- Member of HRH Crown Prince Haakon of Norway's SIKT-network
 - Organised and lead a 100 kilometres rowing expedition in the Hardangerfjord for Oxford rowers
 - Travel the world for food experiences
 - Confident in English and all Scandinavian languages, dabble in French, Spanish and Italian

Peer-Reviewed Publications

Formica, A.F., Burnside, R.J. and Dolman, P.M. (2017) Rainfall validates MODIS-derived NDVI as an index of spatio-temporal variation in green biomass across non-montane semi-arid and arid Central Asia. *Journal of Arid Environments*.

Formica, A.F., Farrer, E.C., Ashton, I.W. and Suding, K.N. (2014) Shrub Expansion Over the Past 62 Years in Rocky Mountain Alpine Tundra: Possible Causes and Consequences. *Arctic, Antarctic, and Alpine Research*.

JOANNA WOLSTENHOLME

39 Pickwick House, George Row, Bermondsey
07967 491950 | Joanna.wolstenholme@gmail.com

EDUCATION

Imperial College London - Science Communication MSc (October 2015 - September 2016) - *Distinction*

- Course covering a mix of academic topics such as philosophy and sociology of science, and more practical experience in journalism, exhibition design, policy writing and many other areas
- Wrote a 10,000 word dissertation looking at how various interest groups – farmers, food traders and scientists – frame the way they talk about GM crops

Jesus College, University of Cambridge - Natural Sciences - specialised into Plant Sciences (2012 – 2015) - *First*

- Achieved a **first** in my final year, and II.i for the previous two years
- Broad biology base, then specialised into Plant Sciences, covering topics such as plant physiology, ecology and global ecosystem modelling, disease spread and genetics
- Confident working from and synthesising primary academic literature

Notre Dame High School Sixth Form, Norwich - International Baccalaureate (2010-2012) – *42 points*

- Results: (subjects in the IB are graded from 1 to 7, with 7 being the highest)
 - Higher level (roughly equivalent to A-level): 7 in Biology, 6 in Chemistry, 6 in English
 - Standard Level (roughly equivalent to AS-level): 7 in Maths, 7 in Philosophy, 6 in Italian
 - Overall score of **42 points**, out of 45 (including a full 3 marks for the core subjects)

Broadland High School, Wroxham, Norfolk (2005-2010)

- 10A*s at GCSE, including English, French, Maths and Sciences, and one A (Art)

RELEVANT EXPERIENCE

Research Manager, [IN-PART](#) (September 2016-present)

- A start-up which catalyses collaboration between academia and industry in the areas of science and technology
- My role is in developing and scaling a new product ([Discover](#)), working closely with the co-founder to run beta tests, secure early clients (mainly large pharma), and develop processes – first for an MVP, and then to scale
- Have gained experience in project and product managing, dealing with corporate stakeholders, assisting sales negotiations, liaising with developers to create a new IT system for the product, co-ordinating and executing large-scale outreach to 100+ universities, information synthesis and report writing

Volunteer Explainer, The Hive at Royal Botanic Gardens, Kew (July 2016 – September 2017)

- Talked to visitors to the Hive (adults and children) about pollination and bees, using my ‘busking bag’ of props
- Have had 5 days of public engagement training with CREW Training

Freelance Blogger, [Ambiental](#) (March 2016 – August 2017)

- Ambiental develop flood risk maps for insurance companies – I wrote about their work in blogs for their website which required me to interview employees about their projects and synthesise existing materials

Intern, Parliamentary and Scientific Committee (February – July 2016)

- Organised a discussion event in Portcullis House on the Zika crisis –Members of both Houses attended

Office of External Affairs, John Innes Centre – Communications Assistant (August and September 2015)

- Drafted press releases, wrote articles for their website and *Science for Africa* newsletter, and developed a media package to explore the fundamental plant science research being carried out at the John Innes

Strategic Initiative in Global Food Security, University of Cambridge – Communications Intern (3 weeks, July 2015)

- Wrote news articles for the website and helped organise and report on the Food Security Symposium

Stellar Leadership Development Program, [British Exploring](#) (August 2015)

- Intensive leadership and self-development program, undertaken whilst canoeing and wildcamping in Norway

ADDITIONAL INFORMATION

- Proficient with InDesign, Word, Excel and PowerPoint, Zoho CRM, and social media tools
- Am a Trustee of the British Exploring Society
- Hold a full, clean driving licence

CURRICULUM VITAE

Jean-Marc BOFFA

Personal information

Born in Paris, France
Married, two children, Jonathan and Jacques

Contact information

The Gems, Apt 12
Borg Oliver Street
Sliema, SLM 1802 Malta
Tel: +356-21310428 (Home); +356-99814828 (Mob.)
Email: j.m.boffa@cqiar.org; jmboffa@melita.com (personal)
Skype: jean.marc.boffa



Education

- **Ph.D., Forestry and Natural Resources, Purdue University**, West Lafayette, IN, USA. 1992-95. Doctoral thesis on the diversity, productivity and management of agroforestry parklands as well as the biophysical shea tree-soil-sorghum interface in Southern Burkina Faso with Drs. D. Knudson, A. Gillespie and P. Coppin. Title: « Productivity and management of agroforestry parklands in the Sudan zone of Burkina Faso, West Africa »
- **M.Sc., Forestry and Natural Resources, Purdue University**. 1989-91. Thesis title: « Land and tree management and tenure policies in Burkina Faso's agroforestry systems »
- **'Ingénieur en agriculture' 5-year degree, Institut Supérieur d'Agriculture, Lille, France**. 1981-86. Thesis work conducted at the Plant and Soil Science Department of the University of Tennessee, Knoxville. Title : « Detection and identification of *Acremonium coenophialum*, a parasitic fungus on tall fescue (*Festuca arundinacea*) affecting livestock performance in Southeastern USA »
- **B.Sc. in Wildlife and Fisheries Biology, University of Massachusetts**, Amherst, MA, USA. 1984-85.

Professional Experience

Director, Terra Sana Projects and Associate Fellow, Science Domain 3 (Tree Diversity, Domestication and Delivery), World Agroforestry Centre (ICRAF), Feb 12-present

Designs and implements research and development activities on systems-oriented, sustainable and climate-smart agriculture for various agencies / programs:

1. Sud Expert Plantes Développement Durable (SEP2D)

Design and implementation of the "Regional innovations network for the genetic improvement and value addition in shea (*Vitellaria paradoxa*)" research project in Benin and Burkina Faso with INRAB, INERA and local partners. Characterization of shea

phenotypic diversity; production and phenological monitoring of phenotypically selected plus trees; development of phenotyping and vegetative propagation methodologies; establishment of *ex situ* collections and village mother orchards for scion production. Inventory of priority improvement traits for value chain actors.

2. Food and Agriculture Organization of the UN, Rome, Italy

Development of a summary booklet on African farming systems for policy makers including trends, potentials and challenges in African agriculture, spatial classification methodology, system overviews, strategic messages and potential applications.

3. World Agroforestry Centre (ICRAF), Nairobi, Kenya

Responsible for the design and implementation of regional germplasm collection strategy in the NSF-funded project « Integrated resource development for the genome-enabled improvement of shea tree for sub-Saharan Africa » led by the University of New Hampshire with ICRAF, CRIG and WACCI. Responsible for partner outreach throughout the shea region, review of current knowledge on shea diversity, inventory of datasets on *ex-situ* collections and phenotypically characterized on-farm populations, selection and field verification of superior shea accessions based on priority morphological, productivity and chemical traits, review and choice of vegetative propagation options and germplasm collection logistics and development of CBD-compatible Memorandi of Understanding with partners toward the establishment of association mapping panels. Development of research resources and tools for shea improvement through the participatory evaluation, propagation and *ex-situ* collection of elite shea accessions in Benin and Burkina Faso in a research proposal to Sud Expert Plantes Développement Durable (SEP2D) scheme of the Institut de Recherche pour le Développement (IRD).

Facilitator of a multidisciplinary research platform for ACIAR-funded project « Farming Systems and Food Security in Sub-Saharan Africa : Priorities for Science and Policy under Global Change » with 15 teams and 65 scientists toward the development of a spatially explicit continental farming systems classification tool for investment planning in Africa. Update of 2001 FAO-World Bank Dixon et al. 'Farming Systems and Poverty' book (www.fao.org/farmingsystems/) including characterization, trends, drivers of change, and strategic priorities for science and policy of 15 major Sub-Saharan farming systems based on expert knowledge, spatial analysis and recent literature. **Book project manager, co-editor and co-author of 600-page book:** Dixon, J., Garrity, D., Boffa, J.-M., Williams, T., Amede, T., with Auricht, C., Lott, R. and Mburathi, G. (eds) 2018. Farming Systems and Food Security in Africa: Priorities for Science and Policy under Global Change. Routledge, London and New York.

Manager of ACIAR-funded project on Strategic applications of farming systems information frameworks for science and policy. Applications of regional SSA farming systems information to CGIAR-CAADP alignment ; FARA's Agriculture Science Agenda for Africa, and Scaling-up Strategy of NEPAD-led African Climate-Smart Agriculture Alliance, May 13-June 14. 1. Organized and facilitated 4-presenter side event on 'Understanding Southern African farming systems : Implications for Policy and Practice' at the Beating Famine Southern Africa Conference, Lilongwe, Malawi, 14-17 April 2015. Production of policy brief on Innovations in farming systems targeting for sustainable intensification and investment impact.

4. Global Shea Alliance (GSA), Accra Ghana.

GSA is the public-private multistakeholder industry association for the shea sector with 400 members from 31 countries including women's groups, food and cosmetic brands and retailers, multinational oil and fat suppliers, research institutions and NGOs.

Comissioned on ICRAF's behalf to provide a review of current knowledge and research needs on opportunities, challenges and risks for shea/karité (*Vitellaria paradoxa*) sustainable management and improvement, as a building block for the development of the Shea Industry's Sustainability Program. Design, organization and facilitation of a regional workshop with shea sector stakeholders for the development of an extension strategy and manual for sustainable shea tree populations, Bamako, 26-28 October 2015. Partnership and project development for GSA's Sustainability Program, July 14-present, including the developmmment of a EuropAid proposal in the call on Inclusive and Sustainable Value Chains and Food Fortification and a fundraising visit to EU and bilateral donors in Brussels, April 2016. Presentations at Annual Shea Conferences, Abidjan, April 2014 ; Ouagadougou, March 2015 ; New York City, June 2015 and Brussels, April 2016.

5. University of Queensland, Brisbane, Australia

Project co-manager, "Farming Systems Approach for Supporting National Investment Planning for Sustainable Intensification and Climate-Smart Agriculture in **Ethiopia**". Review of farming systems information, spatial data analysis, regional workshop and production of a booklet on a national farming systems framework and development priorities for Ethiopia, May 14-May 15.

Project team member, "Piloting a Farming Systems Approach to Investment Planning for Climate-Smart Smallholder Agriculture in **Tanzania**". Farming systems assessment and evaluation process to build local capacity to undertake FS-based planning linking farm-level data to sub-national assessments and investment targeting, June 15-April 16.

Agroforestry systems agronomist, CIRAD, 'SYSTEMS' Research Unit, Yaounde, Cameroon, June 10-December 11

Led the thematic reorientation of the CIRAD Collaborative Research Platform « Pôle de Compétence en Partenariat Grand Sud Cameroun » on agroforestry systems with perennial crops in the Greater Southern Cameroon region as a model of **sustainable intensification of crop production systems** and broadening its partnership base. Organisation of a strategy design workshop, developing the platform scientific framework document in collaboration with reseachers from CIRAD and partner institutions (CIFOR, IITA, ICRAF, IRD, IRAD, and Universities of Yaoundé I and II, Douala, Dschang, Ngaoundéré), coordination and development of a funded US\$500K CORAF/WECARD proposal on productivity and environmental service tradeoffs in Western and Central African cocoa-based agroforestry systems; development of new partnership activities with CIFOR, IITA, ICRAF, and University of Yaoundé II and supervision of student research.

Visiting Researcher/Lecturer, International Environment Institute (now Institute of Earth Systems), University of Malta, Sept 08-May 10

Development of collaborative research projects on sustainable use, quality and origin differentiation as well as multifunctional values of agroforestry Mediterranean fruit tree systems. Development of EU-funded €1.2 M proposal 'Enhancing ecological integrity, connectivity and management effectiveness of protected area landscapes in the Sicily-Malta Cross Border Area (SIMBIOTIC)'. Editing of proceedings of the final WADI project meeting on sustainable management and dynamics of Mediterranean fresh and transitional coastal water bodies. Contribution to MSc. Course 'Toward Sustainable Agriculture' in the Sustainable Environmental Resource Management (SERM) MSc Program jointly offered by James Madison University (Adjunct Assistant Professor status) and the University of Malta.

Consultant for the World Agroforestry Centre (ICRAF)¹ and the Food and Agriculture

Organization of the UN (FAO)², Dec 08-May 10

1) Development of research protocol on biophysical interactions of on farm and on station shaded coffee in Kitale and Ruiru, Kenya, approach for the dissemination of agroforestry techniques within ICRAF's ACIDI-VOCA project in Rwanda, fact sheets on shade species, technical sensitization brochure on agroforestry coffee and participatory learning program for extensionists and pilot farmers in Rwanda;

2) Inception phase for the development of international 2013 FAO Guidelines in Agroforestry for national decision makers (FAO. 2013. Advancing Agroforestry on the Policy Agenda: A guide for decision-makers, by G. Buttoud, in collaboration with O. Ajayi, G. Detlefsen, F. Place & E. Torquebiau. Agroforestry Working Paper no. 1. Food and Agriculture Organization of the United Nations, FAO, Rome. 37 pp.) Analysis of national and global needs, justification, users, objectives, proposed content, development methodology, workplan, establishment of international Steering Committee, and organization of international workshop to discuss modalities for the development of guidelines.

Senior Scientist, Biodiversity, Global Project on Multifunctional Agroforestry Landscapes for Environmental Services, World Agroforestry Centre, Nairobi, Kenya. Aug 05-Aug 08

- Conceptualization, partnership development, resource mobilisation for the development of ICRAF's research programme on enhancing the contribution of agroforestry to landscape-level biodiversity conservation in areas of high global conservation value including:
 - Co-facilitator for the CIFOR-ICRAF Biodiversity Platform aimed at developing approaches and practices for promoting conservation, sustainable use and equitable sharing of biodiversity goods and services in landscape mosaics
 - Promoter of the 'Hotspot Alliance' between Conservation International and ICRAF, to promote the science and practice of agroforestry in landscape management for improved livelihoods, biodiversity conservation and climate change mitigation and adaptation in global Biodiversity Hotspots.
 - Focal point for the ICRAF-CIRAD coffee agroforestry research program on enhancing environmental services and market values of coffee agroforestry systems. Participated with CIRAD, CATIE and University of Wales in organization of 2nd International Symposium on Multi-Strata Agroforestry Systems with Perennial Crops, Turrialba, Costa Rica, 17-21 September 2007.

Senior Scientist (July 02-05) and Scientist (Sept 00-June 02), Tree Domestication and Biodiversity. Country programme manager (July 01-July 05) for ICRAF's research and development programme in Uganda.

- Scientific and administrative leadership in design, planning and implementation of ICRAF's collaborative agroforestry R&D program with emphasis on tree domestication and farm and landscape biodiversity conservation with national and local partners.
- Develop strategies for agroforestry buffer zone interventions around the Mabira Forest Reserve to enhance on-farm tree diversity and increase farmers' incomes based on livelihood and forest resource use surveys with specialized user groups and on-farm tree diversity assessments.
- Develop participatory testing and domestication process for high-value tree species including activities such as rapid market surveys, species priority setting, seed handling research for selected species, assessment of intraspecific diversity and chemical quality of *Prunus africana* in Ugandan forests, germplasm collection and production, and on-farm diversification trials.
- Support dissemination and scaling-up of technology and approaches through partnerships with local governments, development organizations, production of dissemination materials, organization of training and project workshops.

Consultant (Oct. 97- Oct. 99) and interim coordinator (Nov. 99 – May 00) of the Global Forest Genetic Resources Project of the International Plant Genetic Resources Institute, IPGRI (now Bioversity International), Rome.

- Scientific backstopping, liaison and administration of IPGRI forest genetic resources research projects with national partners. Participation in the planning, organisation and coordination of various Project activities, including the first IPGRI-wide FGR Project Meeting.
- Development of concept notes, proposals and project documents for the FGR Project and the establishment of the SSA Programme on Forest Genetic Resources (SAFORGEN). Regional training workshops on conservation and sustainable use of forest genetic resources in West and Central Africa, and Madagascar as well as Eastern and Southern Africa (Ouagadougou, Mar. 1998 and Nairobi, Dec. 1999) and publication of proceedings.

Consultant/author for the Division of Forest Resources, Forestry Department, Food and Agriculture Organization of the United Nations. Preparation of a comprehensive state-of-the-art literature review of agroforestry parkland systems in Sub-Saharan Africa published in English and French in the FAO Conservation Guide Series. Characterization, dynamics, biophysical and socio-economic assets and constraints, management, recommendations for improved management. Author's contract Dec. 96 - Oct. 98 and consultant contracts Dec. 98 and May 00.

Consultant for the Applied Research Component of the Shea Project for Local Conservation and Development, Cooperative Office for Voluntary Organizations (COVOL) Uganda. Appraisal of and input into COVOL's research program for 1999-2002 including biodiversity and *Vitellaria* fruit yield data collection methodologies, April 1999.

Consultant for the European INCO research project on the Improved management of agroforestry parkland systems in Sub-Saharan Africa for the International Program on Arid Land Crops of Ben Gurion University, Beer Sheva, southern Israel. Development of a European INCO-DC proposal in collaboration with the University of Wales, Bangor; technical justification, objectives and work programme; planning and organization of research partnerships including 15 institutional partners in 11 countries, orientation meeting in Beer Sheva. April - Sept. 97. Funded €800K.

Associate Agroforester for the USAID Agricultural Research and Training Support Project within the interdisciplinary Farming Systems Research Program (RSP) of Burkina Faso's agricultural research institute (INERA). Planning, design and implementation of Ph.D farmer-participatory, on-farm research on the diversity, productivity and management of agroforestry parklands as well as tree-crop-soil interactions in Southern Burkina Faso. Support member to the USAID TA team. Supervised 3 research assistants. Oct. 92 - Dec. 93.

Graduate Assistant for the International Programs in Agriculture and the Department of Forestry and Natural Resources at Purdue University. Co-editor of the final technical report for the USAID ARTS project in Burkina Faso, responsible for translations of reports, review and supply of French-English agroforestry and farming systems literature (460 publications), development of two annotated bibliographies on agroforestry, Aug. 90 - Aug 92 and Jan. 94 - Aug. 95.

Visiting Assistant Researcher at the Department of Biology of the University of Malta. Field inventory and mapping of Malta's tree and forest resources. Sampling, identification, GIS

map and data processing on GRASS, data used for the National Resource Management Plan. Feb. - July 89.

Assistant to the United Nations Regional Marine Pollution Emergency Response Center for the Mediterranean Sea, Manoel Island Malta. Technical documents, translations, preparation of training sessions, conferences, library management. Apr. 87 - Aug. 88.

Research Assistant in Plant and Soil Science Department at the University of Tennessee, Knoxville. Detection and identification of a parasitic fungus on tall fescue affecting livestock performance in Southeastern US. Participation in the set-up of an international institutional exchange between UT and ISA. Apr. - Oct. 86.

Professional societies and special responsibilities

International Farming Systems Association; European Agroforestry Federation; Association for Temperate Agroforestry; International Association for Landscape Ecology-Europe
Review of research proposals for the French National Research Agency in 2014, Danish Ministry of Foreign Affairs, the Consultative Research Committee for Development Research in cooperation with the Danish Council for Strategic Research, 2010 and 2011
Member of the Scientific Committee of the Second International Symposium on Multi-strata Agroforestry Systems with Perennial Crops: Making ecosystem services count for farmers, consumers and the environment, 17-21 September 2007, CATIE, Turrialba, Costa Rica
Reviewer for the following journals: Agroforestry Systems; Biodiversity and Conservation; Forests, Trees and Livelihoods, Ethnobiology
Management Committee Member of Neptunes Competitive Swimming Club, Malta, September 2017-present

Languages

French: Native speaker

English: Speaking, reading and writing are **excellent**

Italian: Speaking, reading and writing are good

German: Elementary (high school level)

Spanish: Beginner

Computers: Spreadsheets, databases, graphics, statistics and word processing

Awards and scholarships

2003. Weisheit, A., Bitahwa N. and J-M Boffa. Strengthening capacity of herbalists in the conservation and use of medicinal species in Uganda. **Best Innovative Partnership Program at the 2003 CGIAR Innovation Marketplace**, CGIAR Centers Week, Nairobi, Kenya. \$15,000 to RPWRD as Civil Society partner.

1992 Ph.D Dissertation Research Grant, Midwest Universities Consortium for International Activities, Inc. (MUCIA). \$10,000.

1990-1995, Graduate Assistantship, Purdue University's International Programs in Agriculture\Department of Forestry.

1989-90 Lavoisier Fellowship, French Ministry of Foreign Affairs for Training of French Nationals Abroad.
 1989 Nomination for the J. N. Spaeth and W. R. Boggess Graduate Fellowship in Forestry, Department of Forestry of the University of Illinois at Urbana-Champaign.
 1989 Research Fellowship from the Ministry of Education of Malta within the Franco-Maltese Cultural Agreement.

Selection of organized and facilitated workshops

Atelier de développement d'une stratégie et du contenu de vulgarisation pour la gestion durable des parcs à karité, **ICRAF-Global Shea Alliance-ICCO**, 26-28 octobre 2015, Bamako, Mali
 Expert Group Meeting On Farming Systems In Africa: Context, Issues, Opportunities And Policy Options. **United Nations Economic Commission for Africa** and **ICRAF**, **ICRAF** Headquarters, Nairobi, Kenya, 22 –24 July 2015
 Second Farming Systems Book Update Writeshop, **ICRAF**, Jaccaranda Hotel, Nairobi, 8-10 September 2012
 Workshop for the Review and Development of Farming Systems Chapters. **ICRAF**, Laico Regency Hotel, Nairobi, 25-27 May 2012
 Atelier d'orientation scientifique. Le PCP Agroforesterie Cameroun : plateforme de recherche pour le développement de systèmes agroforestiers durables et performants en Afrique. **CIRAD**, Hôtel Framotel, Kribi, Cameroun, 22-26 novembre 2010
 Development of Agroforestry Guidelines for National Policy and Decision Makers. Inception Workshop. **FAO**, Rome, 7-8 June 2010
 Project Open Day, **ICRAF** Shade Tree Agroforestry Project, Kivu Sun Serena Hotel, Gisenyi, Rwanda, 5 November 2007
 ECOSTAR Project Planning and Partnership Building Workshop, **ICRAF** Bundibugyo Town, 29–30 October 2003
 ECOSTAR Project Design Stakeholder Workshop, **ICRAF**, Bundibugyo District Headquarters, Uganda, 1-2 July 2003
 Fruit tree crops for rural development in Uganda. Uganda tree fruit strategy development workshop, **ICRAF-NARO-UGADEN**, Hotel Africana, Kampala, Uganda, 19 - 21 February 2003
 Priority setting of medicinal tree species in the Lake Victoria lakeshore region, **ICRAF-FORRI** Research Station, Kifu, Uganda, 28 May 2002
 Regional training workshop on conservation and sustainable use of forest genetic resources in Eastern and Southern Africa, **IPGRI**, Nairobi, December 1999
 Atelier régional de formation sur la conservation et l'utilisation durable des ressources génétiques forestières en Afrique de l'Ouest, du Centre et Madagascar, **IPGRI**, Ouagadougou, mars 1998

Selected teaching / training experience

IGCSE Geography lecture, Grade 10 class, Verdala International School, Malta, January 2012
 "Toward sustainable multifunctional agroforestry landscapes". Lecture for ISAT 567, MSc. Course 'Toward Sustainable Agriculture' in UoM and JMU joint Sustainable Environmental Resource Management (SERM) MSc Program, May 2010
 Training module on landscape analysis at the **EcoAgriculture Partners Leadership Course**, Naivasha, Kenya, 12-18 novembre 2006
 "What can Agroforestry provide to the Ugandan farmer? An overview of agroforestry technologies in Uganda". Lecture for training course designed for the **Uganda National Farmers Association**, Kampala, Uganda, March 2005

"Agroforestry for development and biodiversity conservation: Ugandan case studies". Presentation to **European Union Policy Review Team**, Kampala, Uganda, December 2004

"Agroforestry for development and plant diversity conservation: A Ugandan case study and extrapolations". Lecture for the Convention for Biological Diversity - **Global Strategy for Plant Conservation (GSPC) Training course**, Kampala, Uganda, November 2004

"Opportunities and perspectives in Agroforestry: A few highlights of the Agroforestry Research and Development Programme in Uganda". Presentation to the **Household Agricultural Support Programme of DANIDA**, Kampala, Uganda, June 2004

"Concepts and Principles of Tree Domestication". Lecture for the **ICRAF Regional Training-of-Trainers Course in Agroforestry for Eastern and Central Africa**, Makerere University, Kampala, Uganda, 7-23 May 2001

"Tree product quality". Lecture for the **ICRAF Tree domestication training course**, Nairobi, Kenya, 18-24 February 2001

French teacher to first and third year middle-school Maltese students at the **Alliance Française in Malta**. Sept. 88 - June 89.

Academic student supervision

Fahimeh Baziari, **Michigan Technological University**, MSc (2015) The Influence of the Spatial Distribution of Shea Trees (*Vitellaria paradoxa*) on Maize (*Zea mays*) Production in the Upper West Region of Ghana

Véronique Séguy, **AgroParisTech**, 3rd year (2011) Dynamics of farming systems in Central and Southern Cameroon.

Martin Ndzana Eloundou, DEA Sciences économiques, **Univ. Yaoundé II** (2011) Analyses des performances des exploitations cacaoyères innovantes dans la région du Centre Cameroun

Audrey Magné, 2nd year MSc, **ISTOM**, Cergy-Pontoise, France (2008): Agronomic and economic impacts of shade trees in coffee farms in the Lake Kivu region of Rwanda.

Mory Haba, Mémoire Ingénieur, **IRC Montpellier**, France (2008): Diversité et gestion sylvicole dans les systèmes de production des zones caféicoles de Karengé et du Lac Kivu au Rwanda

Grimonie Bernardeau, MSc, **ENSA Montpellier** (2007): Environmental and livelihood functions of trees and their management in coffee agroforestry landscapes of Lake Kivu, Rwanda

Anna Ruotsalainen, MSc, **University of Helsinki** (2007): A cost-benefit analysis of indigenous and exotic agroforestry tree species for enhancing local livelihoods and landscape connectivity in the Taita Hills, Kenya

Lorraine Doherty, BSc, **University of Bangor**, UK (2005) : Sustainable harvesting of medicinal tree species in Bushenyi, Western Uganda

Quentin Meunier, DESS MSc, **University of Paris XII** (2005) : Propagation et conservation d'espèces médicinales prioritaires dans le cadre d'un soutien technique aux tradipraticiens de Rukararwe, Bushenyi district en Ouganda

Julia Schmitt, MSc, **University of Freiburg** (2004) : Impact assessment of watershed extension activities in Katagata, Southwest Uganda

Vincent Bumba, BSc, **Makerere University**, Uganda (2002): Management and use of medicinal plants by communities around Mabira Forest Reserve in Mukono District.

Jean-Gael Jourget, DESS MSc, **University of Paris XII** (2001) : Caractérisation de la diversité arborée dans les systèmes de production agricole dans la zone tampon de la forêt de Mabira en Ouganda du Sud.

Papers and publications

1. Reviews and strategic program documents

- Boffa J.-M. 2015. Opportunities and challenges in the improvement of the shea (*Vitellaria paradoxa*) resource and its management. ICRAF Occasional Paper 24. Nairobi: World Agroforestry Centre.
<http://www.worldagroforestry.org/downloads/Publications/PDFS/B17800.pdf>
Foundation paper used by the Global Shea Alliance to develop the **Sustainability Program of the Shea Industry**.
<http://www.globalshea.com/resources/library/237/GSA-Sustainability-Program>
- Boffa J.-M. 2012. Scientific framework of **CIRAD's** Collaborative Research Platform in Cameroon. 'PCP Agroforestry Cameroon: Research platform for the development of sustainable and productive agroforestry systems in Africa'. CIRAD 46 p.
- Boffa J.-M. 1999. Agroforestry parklands in Sub-Saharan Africa. **FAO Conservation Guide** 34. Food and Agriculture Organization of the United Nations, Rome, Italy. 230 p.
<http://www.fao.org/docrep/005/x3940e/X3940E00.htm>

2. Edited books

- Dixon J., Garrity D., J.-M. Boffa, T. Williams, T. Amede, with Auricht, C., Lott, R. and Mburathi, G. (eds) 2018. **Farming Systems and Food Security in Africa: Priorities for science and policy under global change**. Routledge, London and New York. 600p.

3. Peer-reviewed scientific journal articles

- Boffa, J.M., Hale, I. Lovett, P.N., Danquah, A., Muchugi, A., Dembélé, C., Ramnadass, R. 2018. Improvement of shea (*Vitellaria paradoxa*) in Sub-Saharan Africa: Review of research efforts, characterization data and strategic directions. **Agronomy for Sustainable Development** In progress.
- Pinard F., Boffa J.-M., and Rwakagara E. 2014. Scattered shade trees improve low-input smallholder Arabica coffee productivity in the Northern Lake Kivu region of Rwanda. **Agroforestry Systems** 88 (4) 707-718.
- Ndindeng SA, Talle, Bigoga J, Kengue J, Boffa JM. 2012. Predictors of organoleptic quality of boiled and dried pulp of safou (*Dacryodes edulis*) and the shelf life of its fresh fruits. **Fruits** 67 (2) : 127-136
- Morin A., Bellefontaine R., Meunier Q. and J.-M. Boffa. 2010. Harnessing natural or induced vegetative propagation for tree regeneration in agroecosystems. **Acta Botanica Gallica** 157 (3) 483-492.
- Siriri D, Ong C. K., J. Wilson J, Boffa J.-M and C. R. Black. 2010. Tree species and pruning regime affect crop yield on bench terraces in SW Uganda. **Agroforestry Systems** 78(1):65-77
- Muchugi A. N., Boffa J.-M., Gwali S., Malinga M., Muluvi G. M., Kennedy O. and R. Jamnadass. 2009. Genetic diversity of *Prunus africana* (Hook. f.) Kalkman populations in Uganda. **African Journal of Biotechnology**
- Boffa J.-M., Kindt R., Katumba B., J.-G. Jourget and L. Turyomurugyendo. 2008. Management of tree diversity in agricultural landscapes around Mabira Forest Reserve, Uganda. **African Journal of Ecology** 46:24-32
- Meunier Q., Bellefontaine R., Boffa, J.-M. 2006. Le drageonnage pour la régénération d'espèces médicinales en Afrique tropicale: cas du *Spathodea campanulata* en Ouganda. **Vertigo, La revue électronique en sciences de l'environnement** 7(2): 1-6
<http://www.vertigo.uqam.ca/vol7no2/framerevue.html>
- Boffa J.-M, Turyomurugyendo L, Barnekow-Lillesø JP and R Kindt. 2005. Enhancing farm tree diversity as a means of landscape-based biodiversity conservation: Insights from the Kigezi Highlands, Southwestern Uganda. **Mountain Research and Development** 25(3) 212-217
- Nagawa, F.; Kagorora, K.; Boffa, J.-M.; Owour, C.; Rusoke, C.; Bwengye, E. 2004. Partnership

- building and stakeholder participation in soil erosion management: a case study of Kasitu in Bundibugyo district, Mt Ruwenzori region. **Uganda Journal of Agricultural Sciences** 9: 199-203
- Turyomurugyendo, L.; Boffa, J.-M.; Hakiza, J.J. 2004. Introduction of deciduous fruit tree growing in the tropical highlands of Kabale, Uganda. **Uganda Journal of Agricultural Sciences** 9(1): 470-480
- Katumba, B.M.; Boffa, J.-M.; Abigaba, G.; Okorio, J. 2004. Domestication of medicinal tree species in the Victoria Lakeshore region. **Uganda Journal of Agricultural Sciences** 9: 84-88
- Boffa J.-M, Abigaba G. and Katumba B. 2002. Herbalists and researchers come together to set priorities for the cultivation and conservation of phytomedicinals. **Agroforestry Trends** 3: 15-18.
- Boffa J.-M. 2001. Steps toward indigenous fruit tree domestication. **Agroforestry Trends** 2: 13-14
- Boffa J.-M., S.J.-B. Taonda, J.B. Dickey and D.M. Knudson. 2000. Field-scale influence of the *karité* (*Vitellaria paradoxa*) on white sorghum production in the Sudan zone of Burkina Faso, West Africa. **Agroforestry Systems** 49(2):153-175.
- Boffa J.-M. 2000. West African agroforestry parklands: keys to conservation and sustainable management. **Unasylva** 2000 (51) 11-17.
- Boffa J.-M., G. Yaméogo, P. Nikiéma and J.-B. Taonda. 1996. What future for the shea tree? **Agroforestry Today** 8(4) 5-9.
- Fribourg H.A., Reddick B.B., Boffa J.-M., G. Delaunay G. and R.W. Thompson. 1988. Comparison of microscope stain and ELISA methods to detect *Acremonium coenophialum* in tall fescue. **Tennessee Farm and Home Science** (148) 14-19.

4. Development-oriented and training publications

- Amede T., Auricht C., Boffa J.-M., Dixon J., Mallawaarachchi T., Rukuni M., Teklewold-Deneke T. 2017. A farming system framework for investment planning and priority setting in Ethiopia. **ACIAR Technical Reports Series** No. 92. Australian Centre for International Agricultural Research: Canberra. 52pp.
- Boffa J.-M., J. Dixon, T. Mallawaarachchi and C. Auricht 2016. Farming systems-based investment planning for scaling up climate-smart agriculture in Africa. **Policy Brief** no. tbd, World Agroforestry Centre, 4 p.
- Boffa J.-M., J. Dixon, J. Lynam, C. Auricht, T. Mallawaarachchi and D. Garrity. 2015. Improving the targeting of African farming systems for scaling out sustainable agricultural intensification and enhancing investment impact. **Policy Brief** no. 34, World Agroforestry Centre, 4 p.
- Boffa J.-M., Dixon J., Auricht C. and D. Garrity. 2013. A farming systems framework for targeting investment in Africa. **Policy Brief** no. 16. World Agroforestry Centre, 4 p.
- Lott R., Garrity D., Dixon J., and J.-M. Boffa. 2013. The drivers shaping change in African farming systems. **Policy Brief** no. 17. World Agroforestry Centre, 4 p.
- Boffa. J.-M. 2009. Trees in coffee fields: A **technical brief** for extensionists and coffee farmers in Rwandan and East African highlands. 18 p. Unpublished.
- Pfund J.L, Boffa J.M, Koponen P, O'Connor T, Watts J.D, Colfer C.J.P, Dewi S, van Noordwijk M, Shanley P, Swallow B with colleagues from CIFOR and ICRAF. 2008. Integrating livelihoods and multiple biodiversity values in landscape mosaics. **Research guidelines** version 1: general section, and **Field Methods** http://www.biodiversityplatform.cgiar.org/Documents/LM_Field_Methods_2008.pdf
- Boffa J.-M., Said M. and Pellika. 2007. **Training module** on landscape analysis, EcoAgriculture Partners Leadership Course, Naivasha, Kenya, 12-18 novembre 2006
- Meunier Q., Bellefontaine R., Boffa, J.-M and N. Bitahwa. 2006. Low-cost vegetative propagation of trees and shrubs: **Technical guide** for Ugandan rural communities. CIRAD-ICRAF-Fondation Hulot-RPWRD.
- Tennigkeit T. and J.-M Boffa (eds). 2005. *Maesopsis eminii* for wood production in agroforestry systems in Uganda. Useful Trees for Farming **Technical Bulletin** 4. 30p.

5. Book chapters

- Boffa J.-M., Bagayoko M., Sanders J., Hiernaux P., Ncube S. and J. Nyamangara. 2018. Agro-pastoral Farming Systems of Sub-Saharan Africa. **Book chapter** In : Dixon J., Garrity D., J.-M. Boffa, T. Williams, T. Amede, with Auricht, C., Lott, R. and Mburathi, G. (eds) 2018. Farming Systems and Food Security in Africa: Priorities for science and policy under global change. Routledge, London and New York.
- Garrity D., Dixon J. and J.-M. Boffa. 2017. Understanding African farming systems as a basis for sustainable intensification **Book chapter** In: Oborn I, Vanlauwe B., Phillips M., Thomas R., Brooijmans W. and K. Atta-Krah (eds) 2017. Sustainable Intensification in Smallholder Agriculture: An integrated systems research approach. Routledge, London and New York.
- Auricht C., Dixon J., Boffa J.-M. and D. Garrity, « Farming Systems of Africa », **Book chapter** in: Sebastian K. (ed) 2014. Atlas of African Agriculture Research & Development: Revealing Agriculture's Place in Africa. Washington, D.C.: International Food Policy Research Institute (IFPRI)
- Boffa, J.-M. 2010. Payments for environmental services in Malta's agricultural landscapes. **Book chapter** In E. Conrad and L.F. Cassar (eds.) Perspectives on Landscapes. Institute of Earth Systems, University of Malta: Msida. pp. 65-71.
- Pfund J.-L., Koponen P, O'Connor T, Boffa J.-M., Meine van Noordwijk M. and J.-P. Sorg. 2008. Biodiversity conservation and sustainable livelihoods in tropical forest landscapes. **Book chapter** In: Laforteza R., Chen J., Sanesi G. and T. R. Crow (eds.). Patterns and Processes in Forest Landscapes - Multiple Use and Sustainable Management. Springer, Netherlands. pp 297-322.
- Swallow, B., Boffa, J.-M.; Scherr, S.J. 2006. The potential for agroforestry to contribute to the conservation and enhancement of landscape biodiversity. **Book chapter** In: Garrity DP, Okono A, Grayson M and Parrott S, (eds.). World Agroforestry into the Future. Nairobi : World Agroforestry Centre (ICRAF) pp. 95-101.

6. Workshop and conferences papers

- Boffa J.-M. and A. Kalinganire. 2016. Shea research and development activities in the Sahel. Presentation at the Shea European Trade & Industry Conference, Brussels, Belgium, 11 April 2016.
- Mishili F., Mallawaarachchi T., Valerian J., Auricht C., Boffa J.-M. and J. Dixon. 2016. **Contributed presentation** at the 60th Australian Agricultural and Resource Economics Society (AARES) Annual Conference, Canberra, ACT, 2-5 February 2016.
- Boffa J.-M. 2015. Opportunities and challenges for the sustainable management of shea parklands. **Presentation** at Atelier de développement d'une stratégie et du contenu de vulgarisation pour la gestion durable des parcs à karité, ICRAF-GSA-ICCO, 26-28 octobre 2015, Bamako, Mali.
- Boffa J.-M. 2015. Regional and national farming systems information for planning: Background & approach. **Presentation** at Workshop 'Piloting a Farming Systems Approach to Investment Planning for Climate-Smart Smallholder Agriculture in Africa', Sokoine University of Agriculture, University of Queensland, 16-17 Sept 2015, Nashera Hotel, Morogoro, Tanzania.
- Boffa J.-M. 2015. Sustaining diverse and productive shea parklands. **Presentation** at North American Shea Trade and Industry Conference, 11 May 2015, Radisson Martinique Hotel, New York
- Boffa J.-M., C. Auricht, J. Dixon and D. Garrity. 2015. A Farming Systems Approach to Support Planning and Investment in Agriculture. **Presentation** at Beating Famine Southern Africa Conference : Sustainable Food Security through Land Regeneration in a Changing Climate, 14-17 April 2015, Bingu International Conference Centre, Lilongwe, Malawi.

- Boffa J.-M. 2015. Opportunities and challenges for the improved and sustainable management of shea parklands. **Presentation** at Shea 2015 : A Promising Future, Ouagadougou, Burkina Faso, 23-25 March 2015.
- Boffa J.-M. 2015. Farming Systems Project and Approach. **Presentation** at Workshop 'Ethiopian Farming Systems Information for Planning Investment in Agricultural Development', ICRAF, University of Queensland, ILRI campus, Addis Ababa, Ethiopia, 24-25 February 2015.
- Boffa J.-M. 2014. Regenerating shea parklands for sustained productivity and diversity: Strategy components. **Presentation** at Shea 2014 ; The industry unites, Abidjan, Côte d'Ivoire, 24-26 March 2014.
- Garrity D., J. Dixon and J.-M. Boffa. 2012. Understanding African farming systems : Science and policy implications. **Paper** prepared for the Conference 'Food security in Africa : Bridging research and practice', Sydney, 29-30 November 2012. ACIAR / Australian International Food Security Center, 63 p.
- Ndzana Eloundou M.J., J.M. Boffa, R. Bourgoing, C.B. Kamdem, B. Kamgnia Dia and H. Todem. 2012. Analyse de la rentabilité et de l'efficacité technique des systèmes agroforestiers cacaoyers innovants dans la région du centre Cameroun. **Paper** presented to the 17th International Cocoa Research Conference, Improving the profitability of small and medium-sized farms : The principal key to a global sustainable cocoa economy. COPAL and Government of Cameroon, Yaounde, Cameroon, 15-20 October 2012.
- Boffa J.-M., Pinard F., Kagoro D., Ngendo M., Rwakagara E. and S. Franzel. 2009. Tree shade and coffee interactions in the northern Lake Kivu region, Rwanda. **Poster** presented at the Second World Agroforestry Congress: The Future of Global Land Use, Nairobi 23-28 August 2009
- Scapini F., Boffa J.-M., F. Cassar L.F. , Conrad E. and M. Nardi (**Editors**). 2009. Sustainable management of Mediterranean coastal fresh and transitional water bodies: a socio-economic and environmental analysis of changes and trends to enhance and sustain stakeholder benefits. **Proceedings** of the international conference of the WADI Project (INCO-CT2005 015226), Malta, 5-8 November 2008
- Kindt, R., Boffa, J.M., Lengkeek, A.G., Degrande, A., Mbosso, C. and Van Oijen, D. 2008. Tree species diversity in cultivated landscapes: Investigation of biodiversity patterns through training of African researchers. In: Temu AB, Chamshama SAO, Kung'u J, Kaboggoza J, Chikamai B and Kiwia A. (eds.) New Perspectives in Forestry Education. Peer reviewed **paper** presented at the First Global Workshop on Forestry Education, September 2007. ICRAF, Nairobi, Kenya, pp. 384-400.
- Boffa J.-M., Pinard F., Kagoro D., Ngendo M., Rwakagara E. and S. Franzel. 2007. Tree shade and coffee interactions in the northern Lake Kivu region, Rwanda. **Paper** presented at Second International Symposium on Multistrata Agroforestry Systems with Perennial Crops, Turrialba, 17-21 September 2007
- Pfund, J.L., O'Connor T.; Koponen P. and J.-M. Boffa. 2006. Transdisciplinary research to promote biodiversity conservation and enhanced management of tropical landscape mosaics. **Paper** In : Laforteza R. and Sanesi G. (eds.) 2006. Patterns and processes in forest landscapes. Consequences of human management. Proceedings of the 4th Meeting of IUFRO Working Party 8.01.03, Sept 26-29, 2006, Locorotondo, Bari, Italy, pp. 139-148.
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7. Working papers

Villamor G. B., Ambal R. G., Boffa J.-M., Brooks T., Cereno A., Coroza O., Lasco R. D., Schroth G. and M. van Weerd. 2008. Setting landscape conservation targets and promoting them through compatible land use in the Philippines - a meeting-workshop of CI-ICRAF Collaborative Project. **ICRAF Working Paper** no. 74. Bogor, Indonesia. World Agroforestry Centre.

Selected Resource Mobilization and R&D Project Management Experience

Project name:	Study on Shea Butter Value Chain in Mali
Funding scheme:	World Bank eConsultancy Services
Dates:	December 2017
Partners:	Terra Sana Projects, World Agroforestry Centre (ICRAF) and Association Malienne des Professionnels du Karité (APROKAM)
Budget:	\$50k, Expression of Interest shortlisted with 6 other organizations
Role:	Proposal development leader
Project name:	Réseau régional d'innovations pour l'amélioration génétique et la valorisation du karité (KARITE-PLUS)
Funding scheme:	Sud Expert Plantes Développement Durable (SEP2D) Call. Donors : AFD, FFEM, Ministère des Affaires Etrangères, IRD, MNHN
Dates:	November 2017 - October 2020
Partners:	INRAB, Université d'Abomey-Calavi, Natura, FLUDOR, Benin; INERA, RPBHC, Olvea, Burkina Faso; Terra Sana Projects, Malta and Global Shea Alliance.
Budget:	€80K. Funded; 6 funded projects out of 166 submitted proposals.
Role:	Proposal development leader
Project name:	Strengthening Ghana's shea value chain through resource sustainability, producer capacity and market access innovations
Funding scheme:	European Commission, EuropeAid/151093/DH/ACT/Multi, Call on Inclusive and Sustainable Value Chains and Food Fortification
Dates:	September 2016 - August 2021
Partners:	Global Shea Alliance, ICRAF, ICCO, MEDA, ITC, CRIG
Budget:	€7M (€4.9M EU, €2.1M other) graded 43.5/50. Not funded.
Role:	Technical assistant to proposal leader (Global Shea Alliance)
Project name:	Integrated resource development for the genome-enabled improvement of shea tree for sub-Saharan Africa
Funding scheme:	Bill and Melinda Gates Foundation through the US National Science Foundation's BREAD and ABRDC programs - Basic Research to Enable Agricultural Development and Advancing Basic Research in Crop Plants Relevant to Smallholder Agriculture in Developing Countries
Dates:	May 2016 – April 2019
Partners:	University of New Hampshire, ICRAF, CRIG, WACCI
Budget:	\$563K. Funded.
Role:	Wrote the proposal's rationale and justification; team member

Project name: Farming Systems and Food Security in Africa: Strategic framework for utilizing outcomes for science and policy applications
Funding scheme: Australian Center for International Agricultural Research (ACIAR)
Dates: May 2013-April 2014
Partners: NEPAD, FARA, African Climate-Smart Agriculture Alliance
Budget: \$186K. Funded.
Role: **Participant in proposal development; ICRAF project manager**

Project name: Farming Systems and Food security in Africa: Priorities for Science and Policy under Global Change
Funding scheme: Australian Center for International Agricultural Research (ACIAR)
Dates: May 2012-June 2013
Partners: 64 authors from different institutions
Budget: \$320K. Funded.
Role: **Participant in proposal development; ICRAF project manager**

Project name: Optimizing productivity and perennial intercrop diversity tradeoffs in West and Central African cocoa farms
Funding scheme: West and Central African Council for Agricultural Research and Development (CORAF/WECARD)
Dates: November 2011-October 2014
Partners: IRAD, CIRAD, Universities of Dschang and Yaoundé I, ICRAF-Cameroun; KNUST in Ghana; CNRA and University of Cocody, in Côte d'Ivoire
Budget: \$500K. Funded.
Role: **Proposal development leader**

Project name: SIMBIOTIC: Enhancing Sicily-Malta BIOgeographical Transboundary Insular Connectivity
Funding scheme: European Union Italy-Malta Cross-Border Cooperation Programme
Dates: November 2010- December 2013
Partners: Earth Systems Institute, U. of Malta; Ministry for Gozo; Dip. di Botanica dell'Università degli Studi di Catania; Comune di Pachino
Budget: €1.2M. Funded.
Role: **Proposal development leader**

Project Name Biodiversity conservation and sustainable livelihoods through forest corridor implementation in Taita Hills, Kenya.
Funding scheme European Commission - EuropeAid
Project date 48 months (2009-2013)
Partners Univ of Helsinki, Univ of Ghent, Kenya Forest Service, World Agroforestry Centre, National Museum of Kenya, East African Wildlife Society, Taita Taveta Wildlife Forum
Budget Total \$2.1M; ICRAF component of \$210K. Not funded.
Role **ICRAF Focal point for proposal development.**

Project Name BioArk: Capturing biodiversity values and enhancing livelihoods in multifunctional landscapes of the Eastern Arc Mountains
Funding scheme European Commission - EU-FP7-ENV-2007-1-2.1.4.3. Biodiversity values, sustainable use and livelihoods.
Project date 48 months (2008-2012)
Partners Univ of Helsinki, Univ of Ghent, Univ of Cambridge, World Agroforestry Centre, National Museums of Kenya, East African Wildlife Society, Taita

Budget	Taveta Wildlife Forum, Tanzania Forest Conservation Group, University of Sokoine
Role	Total \$3.5M; ICRAF component of \$700K. Ranked 13.5/15. Not funded. ICRAF Focal point for proposal development.
Project Name	Integrating Livelihoods and Multiple Biodiversity Values in Landscape Mosaics.
Funding scheme	Swiss Agency for Development and Cooperation
Project date	June 07 – May 10
Partners	CIFOR-ICRAF Biodiversity Platform
Budget	Total \$1.0M; ICRAF component of \$290K. Funded.
Role	ICRAF Focal point for proposal development and project manager
Project Name	CAFNET: Connecting, enhancing and sustaining environmental services and market values of coffee agroforestry in Central America, East Africa and India.
Funding scheme	EuropeAid - Programme on Environment in Developing Countries
Project date	January 07 - December 10
Partners	CIRAD, World Agroforestry Centre (ICRAF), University of Wales, Bangalore University, CATIE, Coffee Board of India.
Budget	Total \$3.14M, ICRAF component of \$625K. Funded.
Role	Participant in proposal development; ICRAF project manager
Project Name	Tree Shade Agroforestry Project: Promoting agroforestry to enhance quality, diversification and sustainability of smallholder coffee production systems in the Gisenyi areas of Lake Kivu.
Funding scheme	ACDI-VOCA/USAID
Project date	May 06 - August 08
Partners	COOPAC
Budget	\$488K. Funded.
Role	Participant in proposal development; ICRAF project manager
Project Name	Assessment of bark quality and genetic diversity in Ugandan <i>Prunus africana</i> populations for the development of on-farm cultivation and export markets.
Funding scheme	USAID - Investment in the Development of Export Agriculture Project, Chemonics International
Project date	January 02 - December 03
Partners	FORRI
Budget	\$85K. Funded.
Role	Proposal development leader; ICRAF project manager
Project Name	ECOSTAR: Empowering Communities for Sustainable Agriculture and Rural Development in Bundibugyo District, Uganda
Funding scheme	DANIDA-Household Agricultural Support Programme
Project date	May 03 - June 04
Partners	Bundibugyo district
Budget	\$270K. Funded.
Role	Proposal development leader; ICRAF project manager

Project Name	Agroforestry Research and Development to conserve and enhance biodiversity in the Kigezi Highlands and Mabira forest buffer zone of Uganda.
Funding scheme	USAID
Project date	September 99 - April 04
Partners	FORRI, Kampala and Kabale sites
Budget	\$2.7M. Funded.
Role	ICRAF Project manager (June 00-April 04)

Project Name	Accelerating the diversification of agroforestry systems for improved livelihoods and biodiversity conservation in Southwestern Uganda.
Funding scheme	USAID - The Environmental Conservation Trust of Uganda (ECOTRUST)
Project date	October 02 - June 04
Partners	Rukararwe Partnership for Rural Development
Budget	\$100K. Funded.
Role	ICRAF Project manager
